

BFS Traversal

Step 1: Start at A

Legend

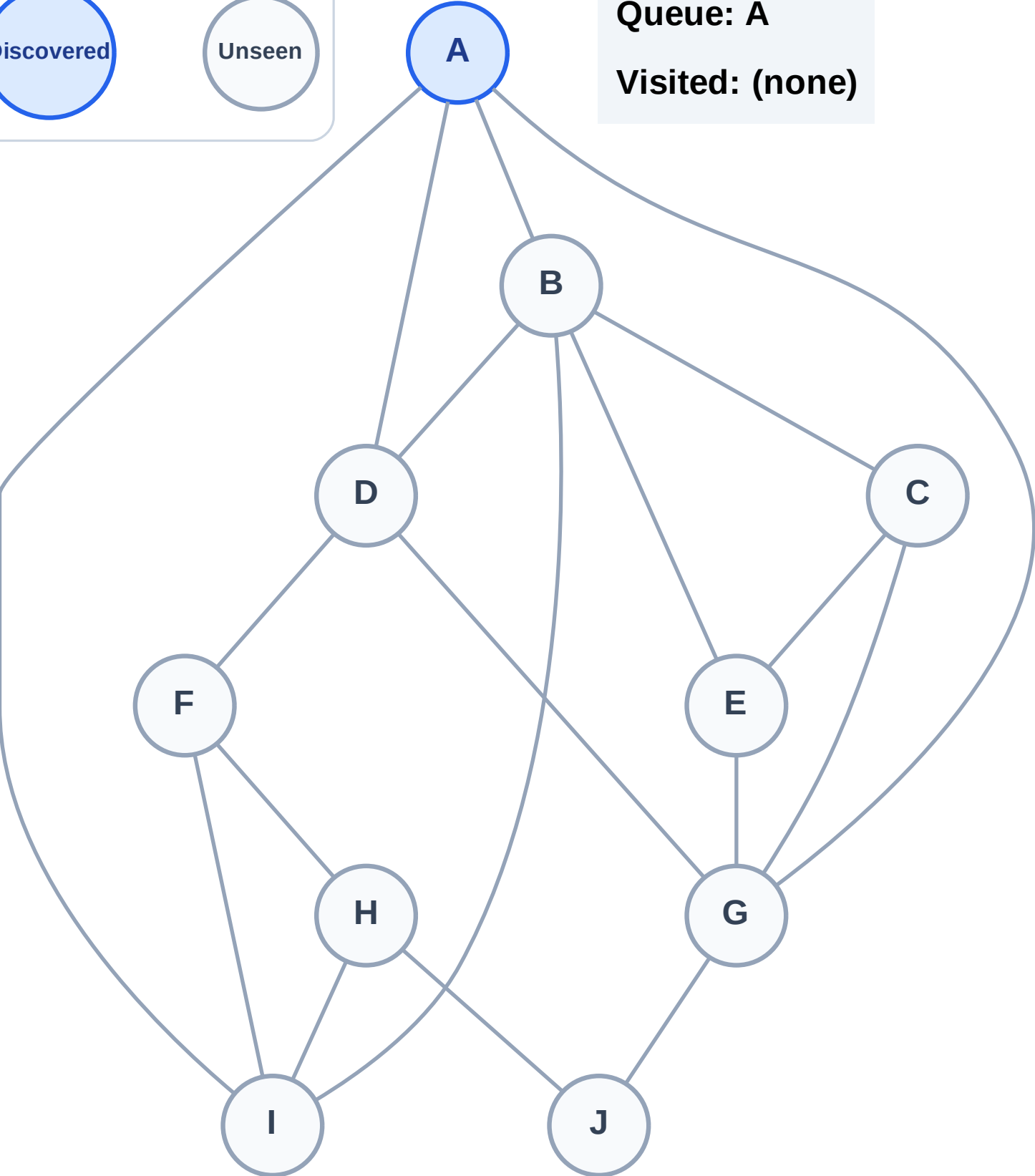
Complete

Processing

Discovered

Unseen

Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

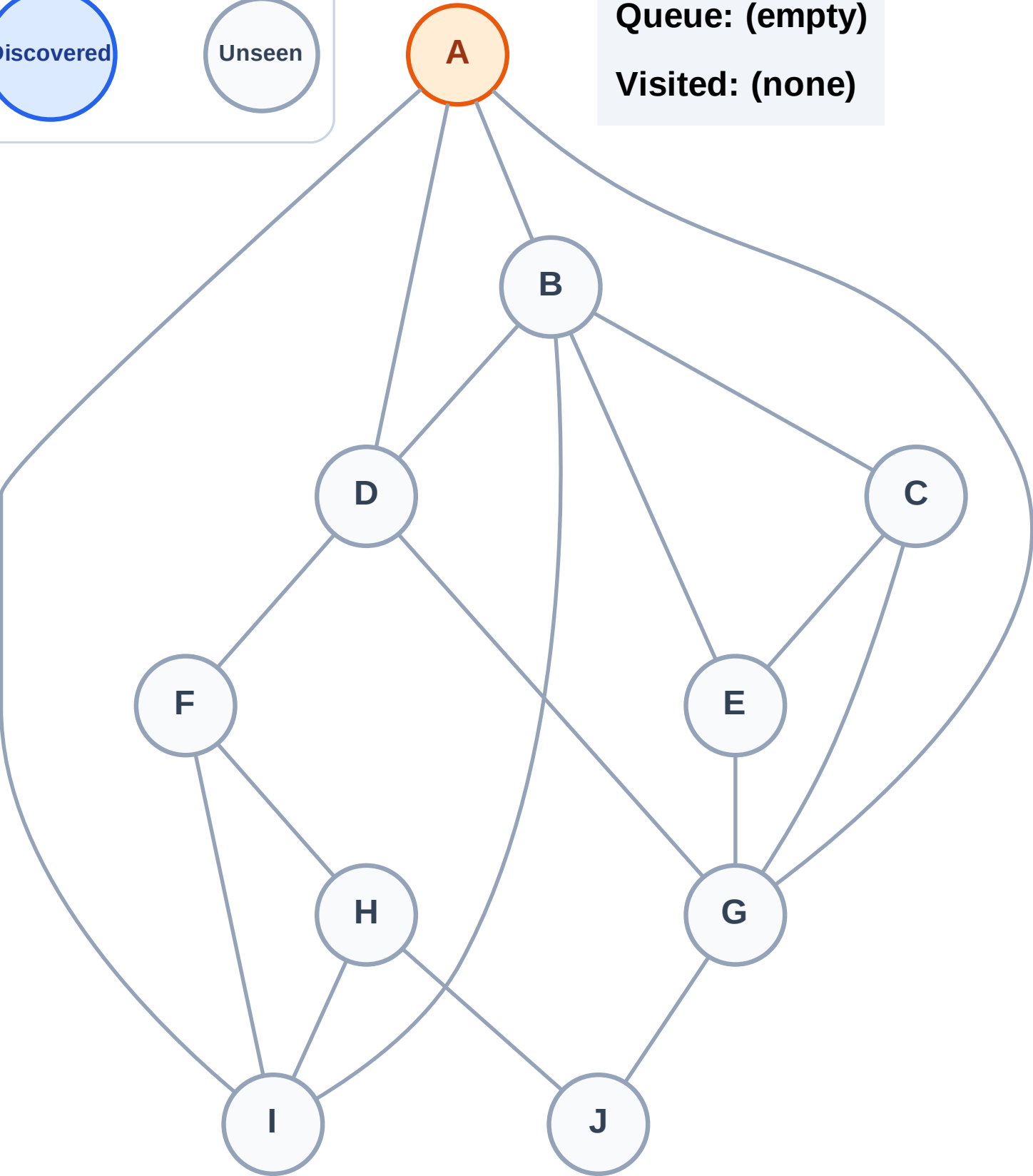
Complete

Processing

Discovered

Unseen

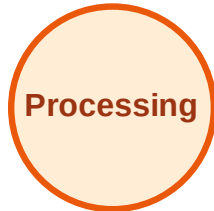
Queue: (empty)
Visited: (none)



BFS Traversal

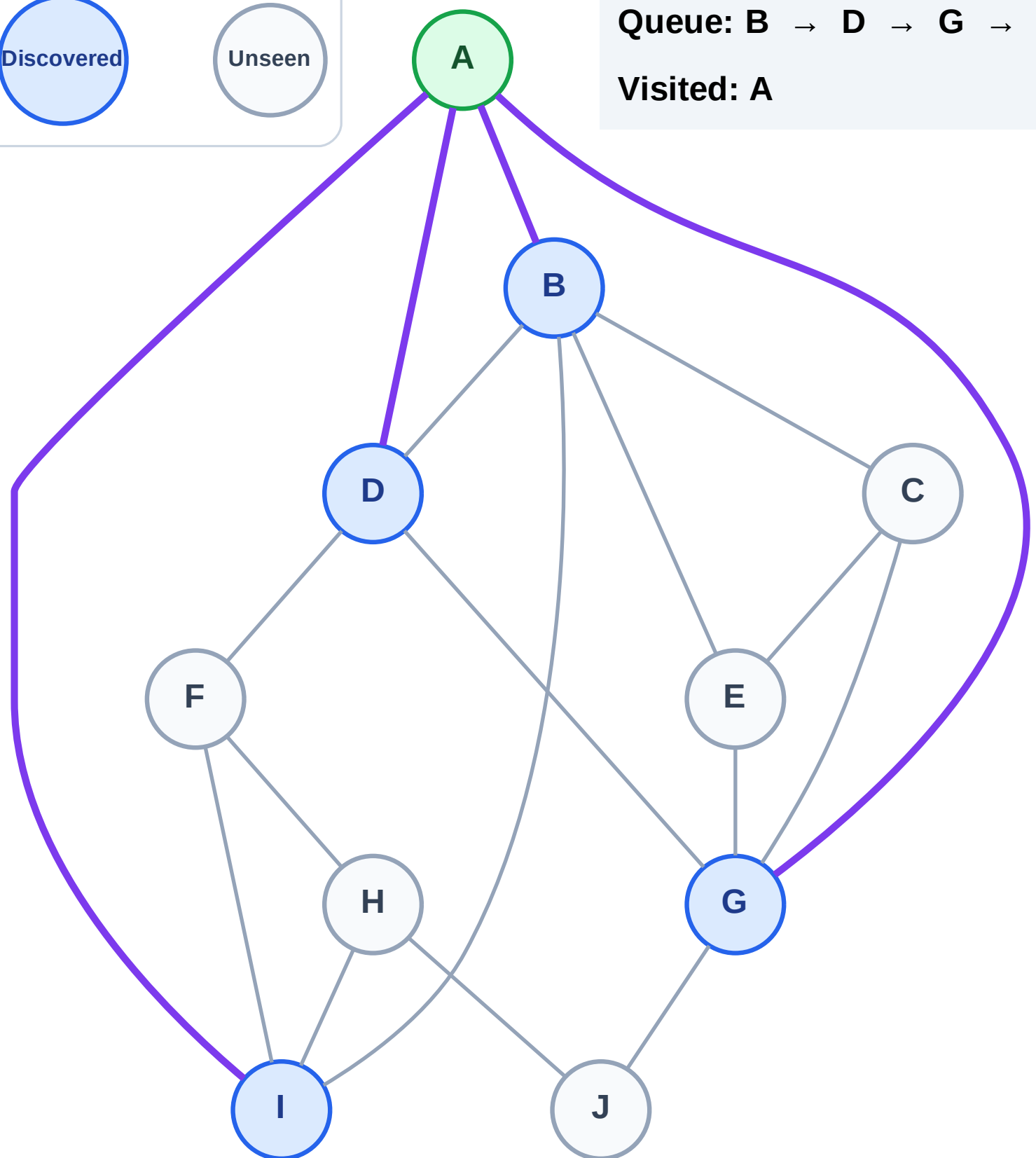
Step 3: Discover B, D, G, I from A

Legend



Queue: B → D → G → I

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

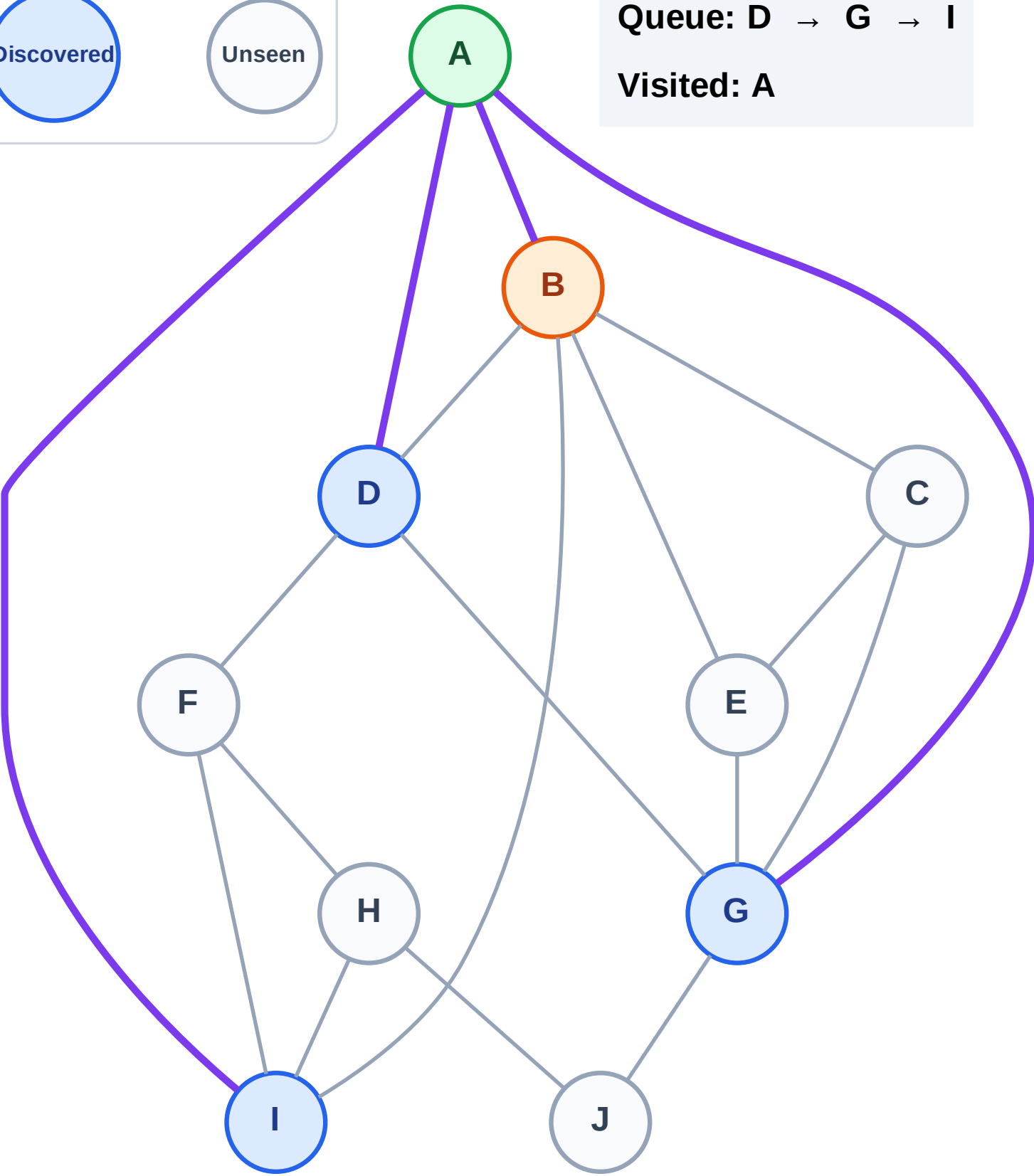
Processing

Discovered

Unseen

Queue: D → G → I

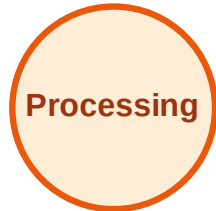
Visited: A



BFS Traversal

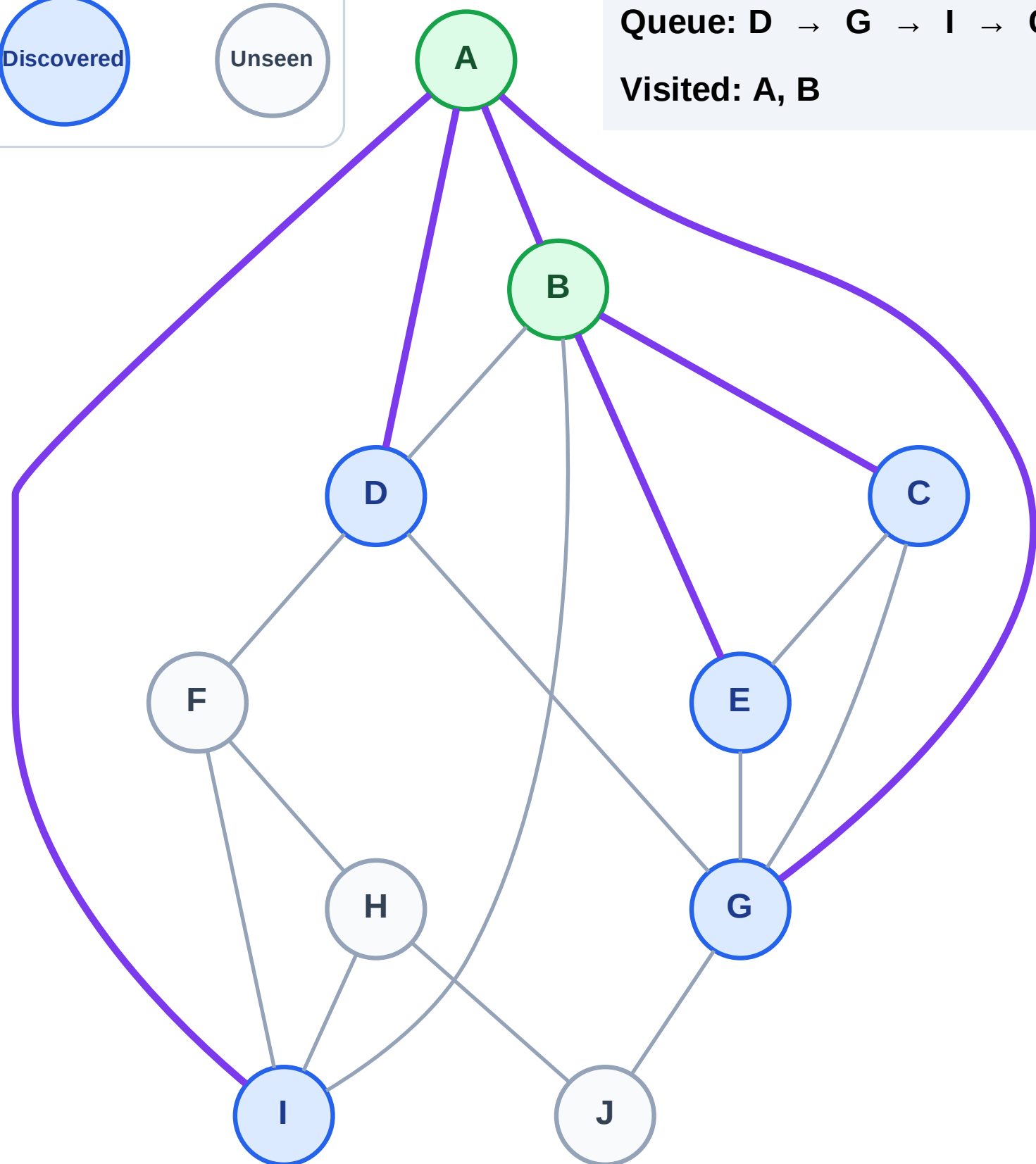
Step 5: Discover C, E from B

Legend



Queue: D → G → I → C → E

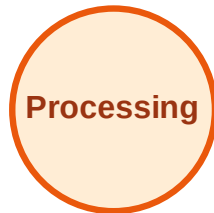
Visited: A, B



BFS Traversal

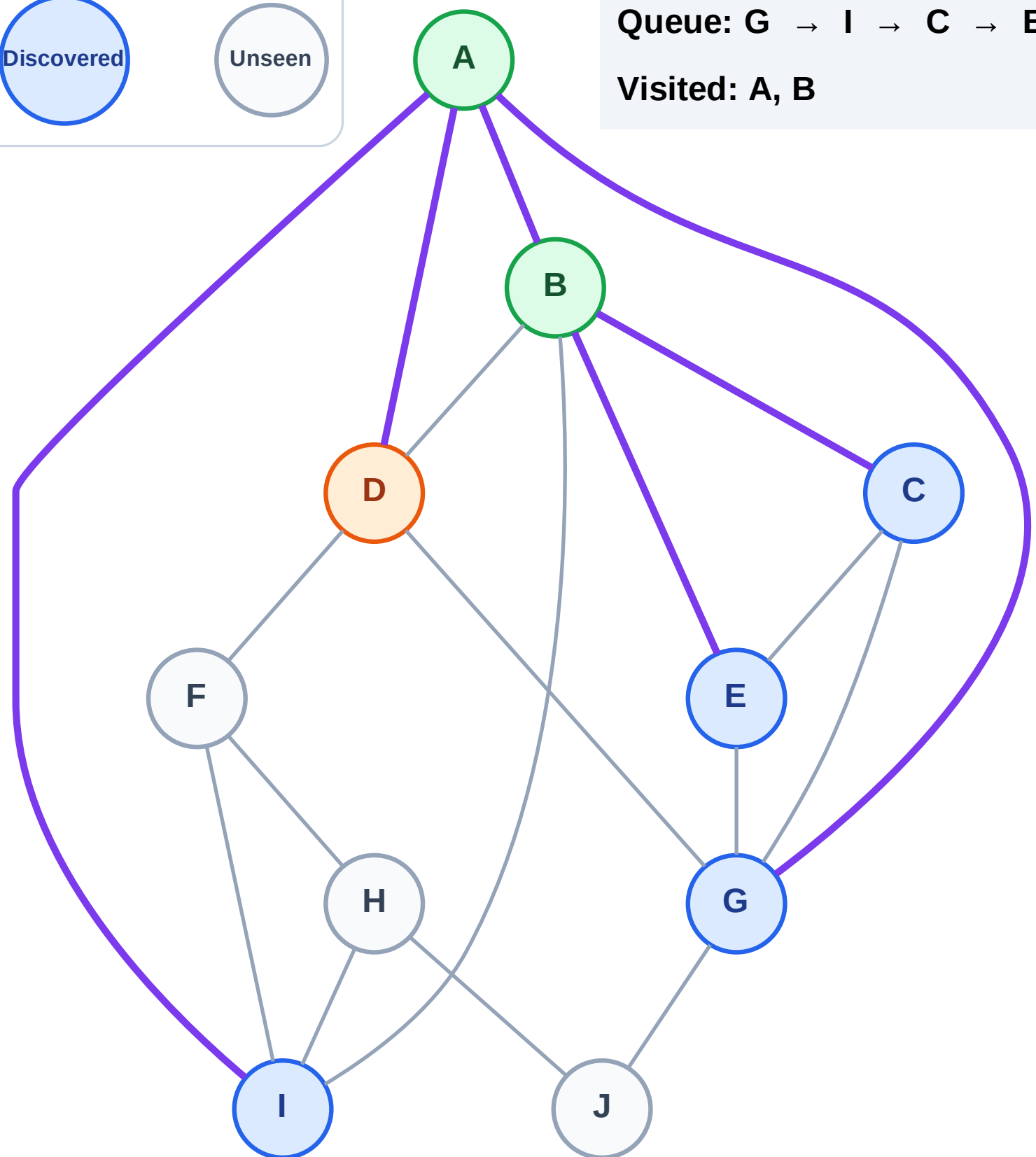
Step 6: Process node D

Legend



Queue: G → I → C → E

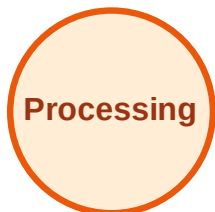
Visited: A, B



BFS Traversal

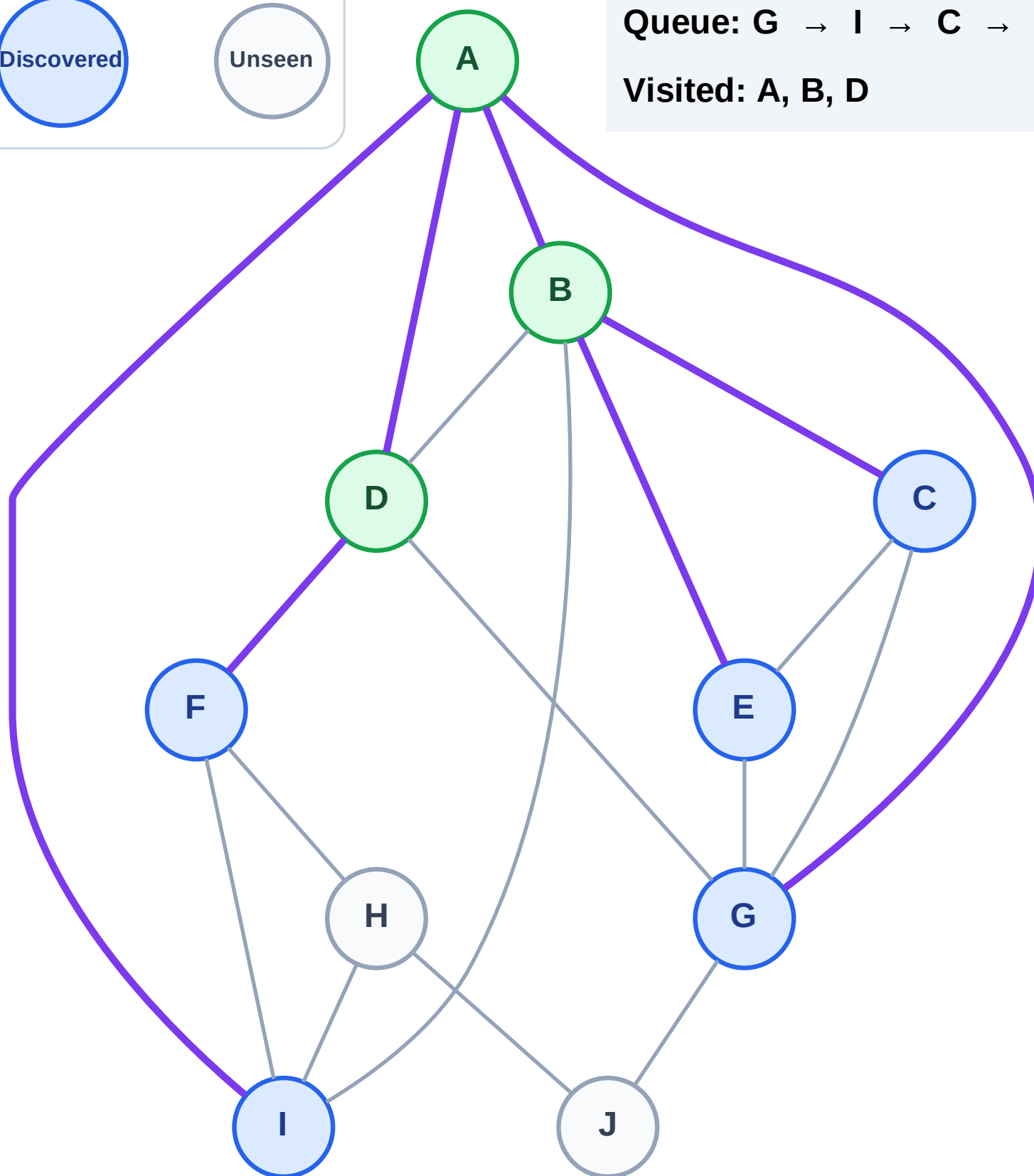
Step 7: Discover F from D

Legend



Queue: G → I → C → E → F

Visited: A, B, D



BFS Traversal

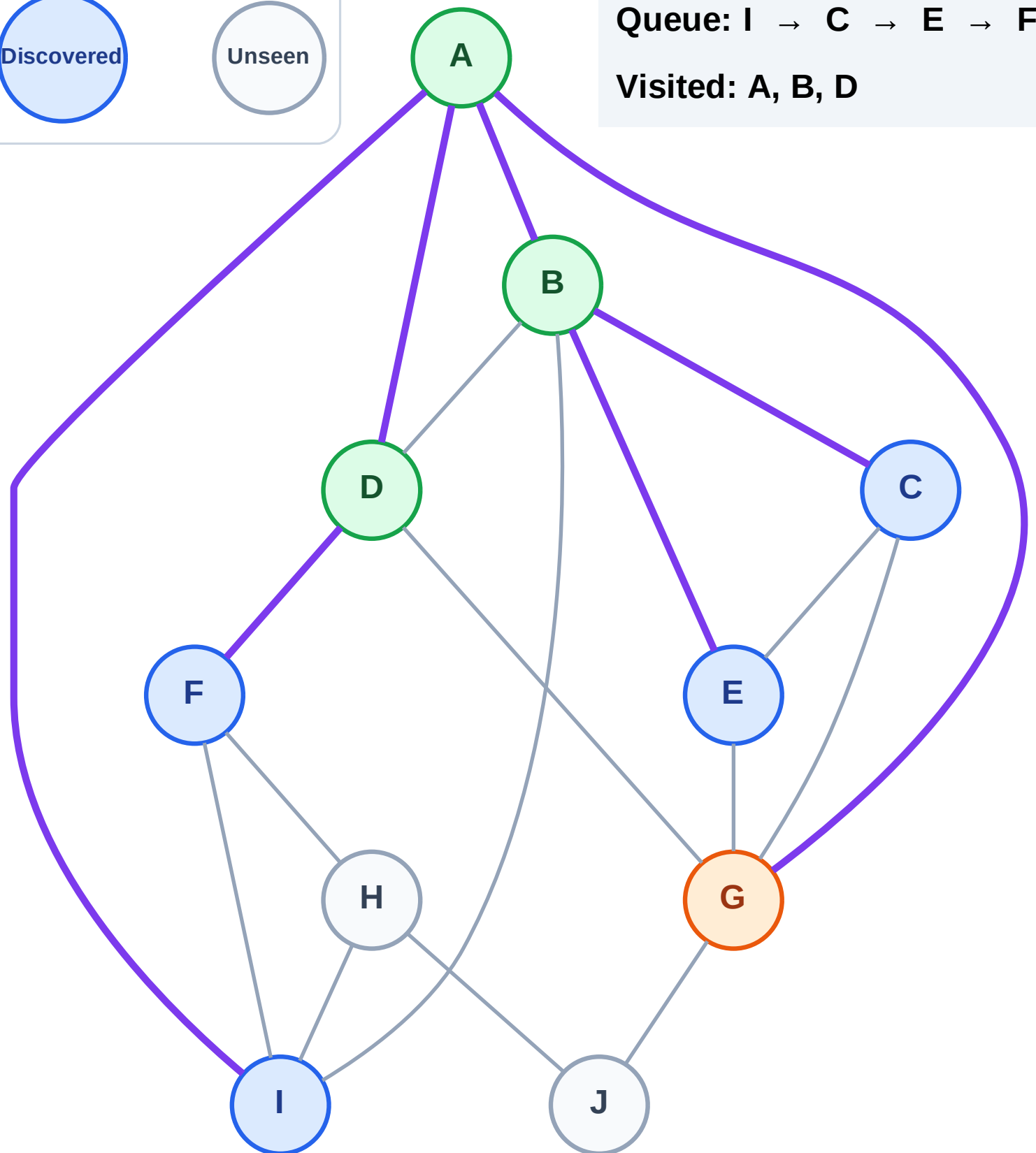
Step 8: Process node G

Legend



Queue: I → C → E → F

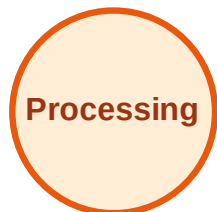
Visited: A, B, D



BFS Traversal

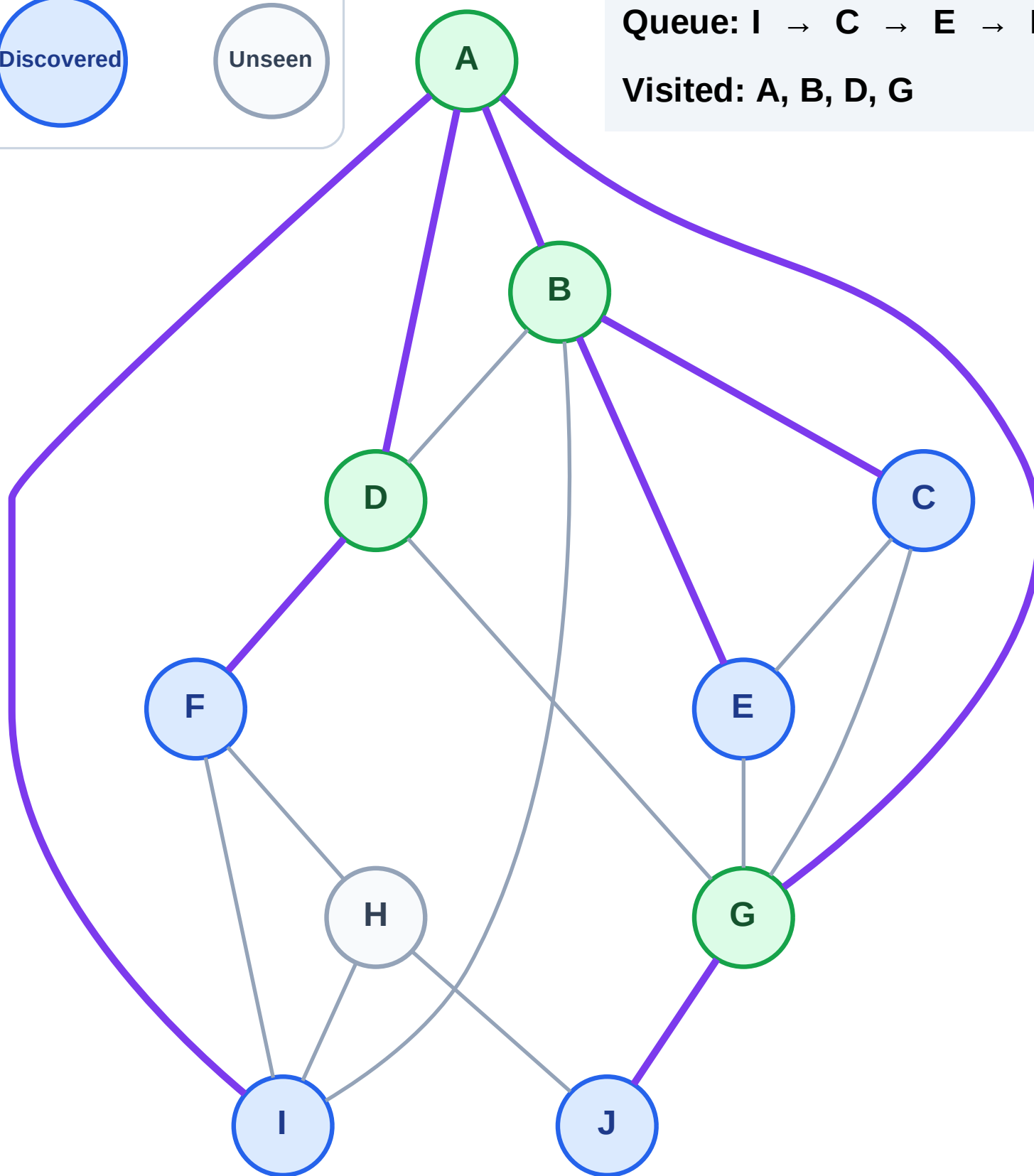
Step 9: Discover J from G

Legend



Queue: I → C → E → F → J

Visited: A, B, D, G



BFS Traversal

Step 10: Process node I

Legend

Complete

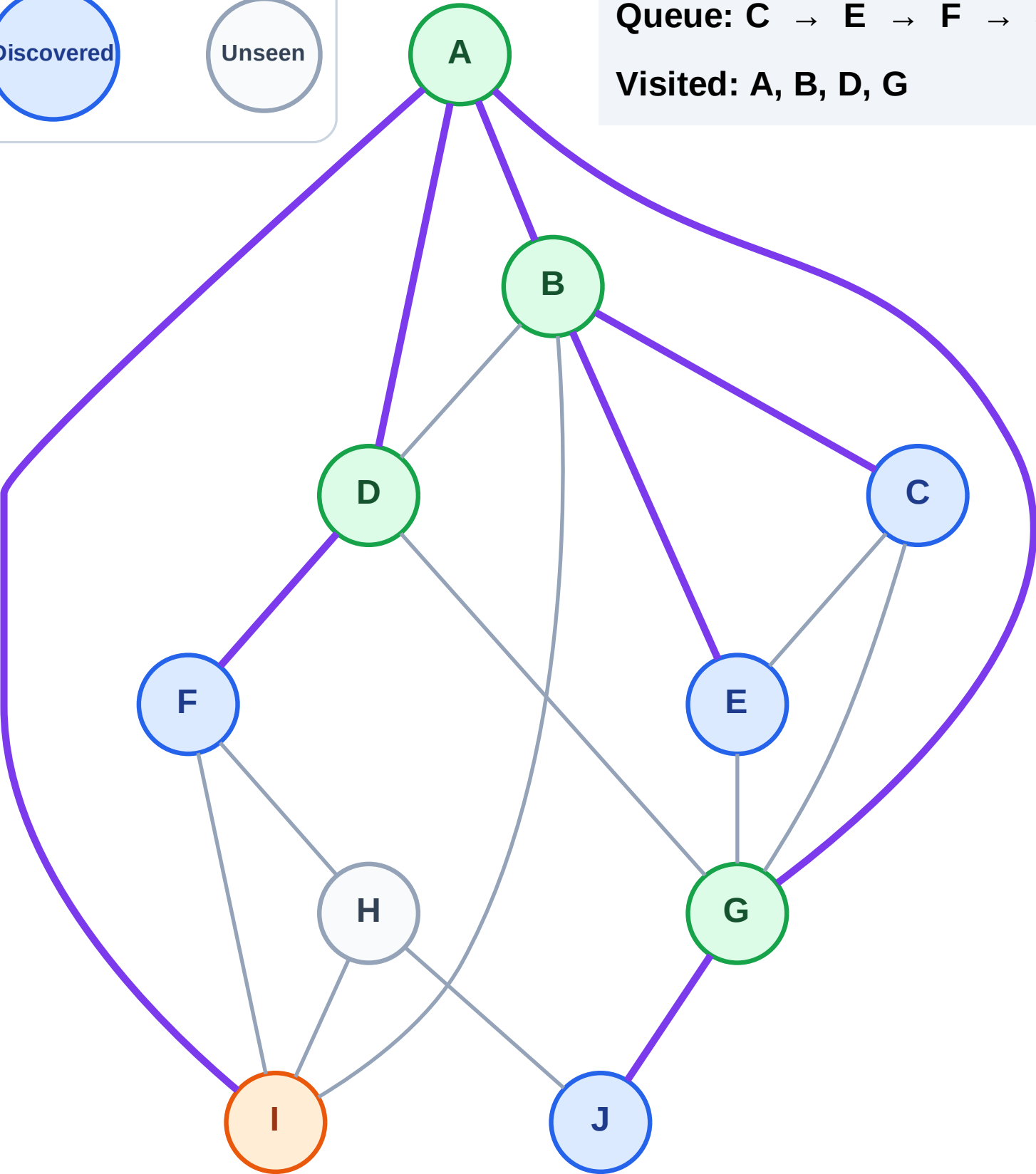
Processing

Discovered

Unseen

Queue: C → E → F → J

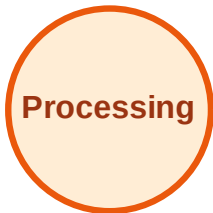
Visited: A, B, D, G



BFS Traversal

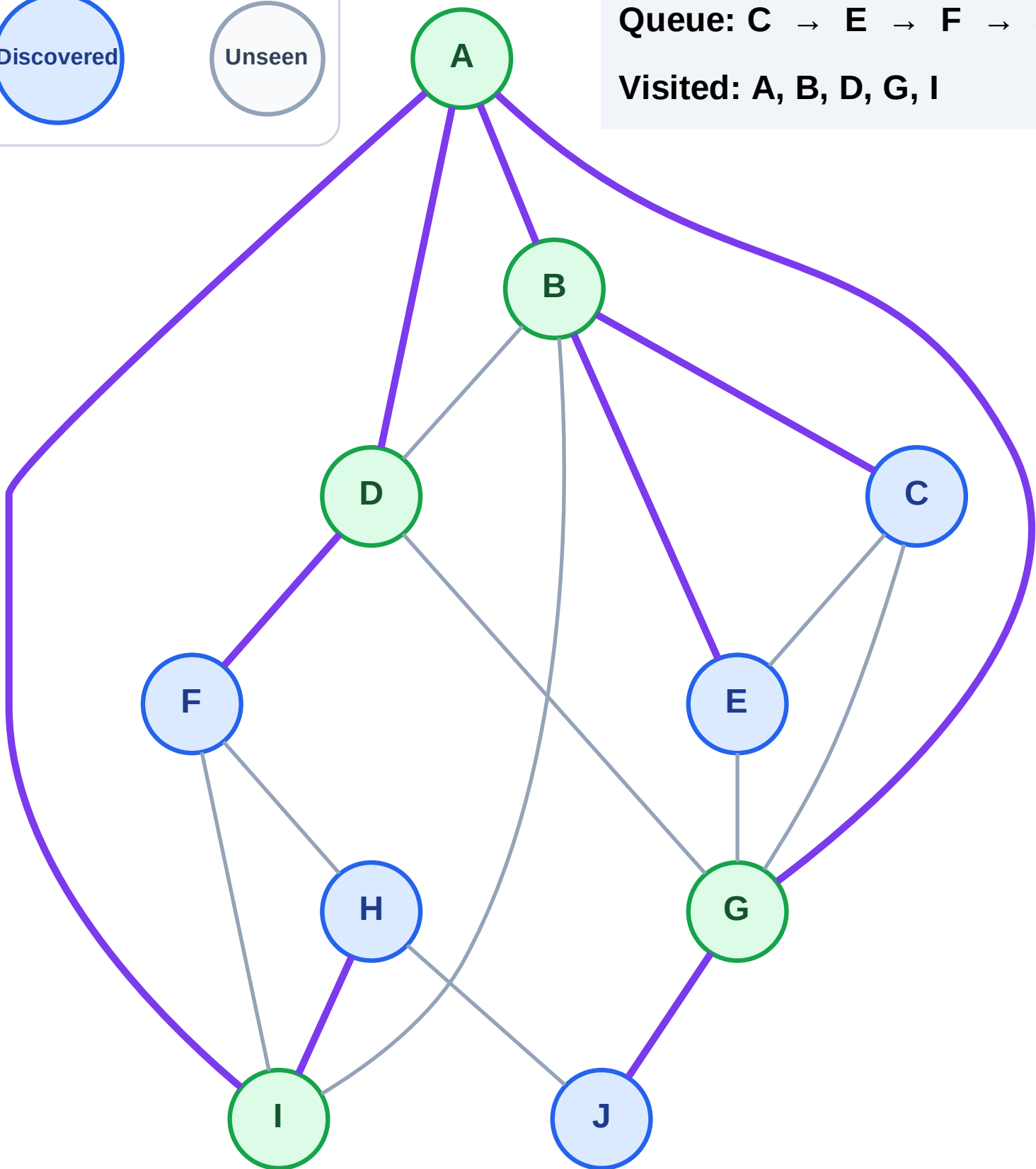
Step 11: Discover H from I

Legend



Queue: C → E → F → J → H

Visited: A, B, D, G, I



BFS Traversal

Step 12: Process node C

Legend

Complete

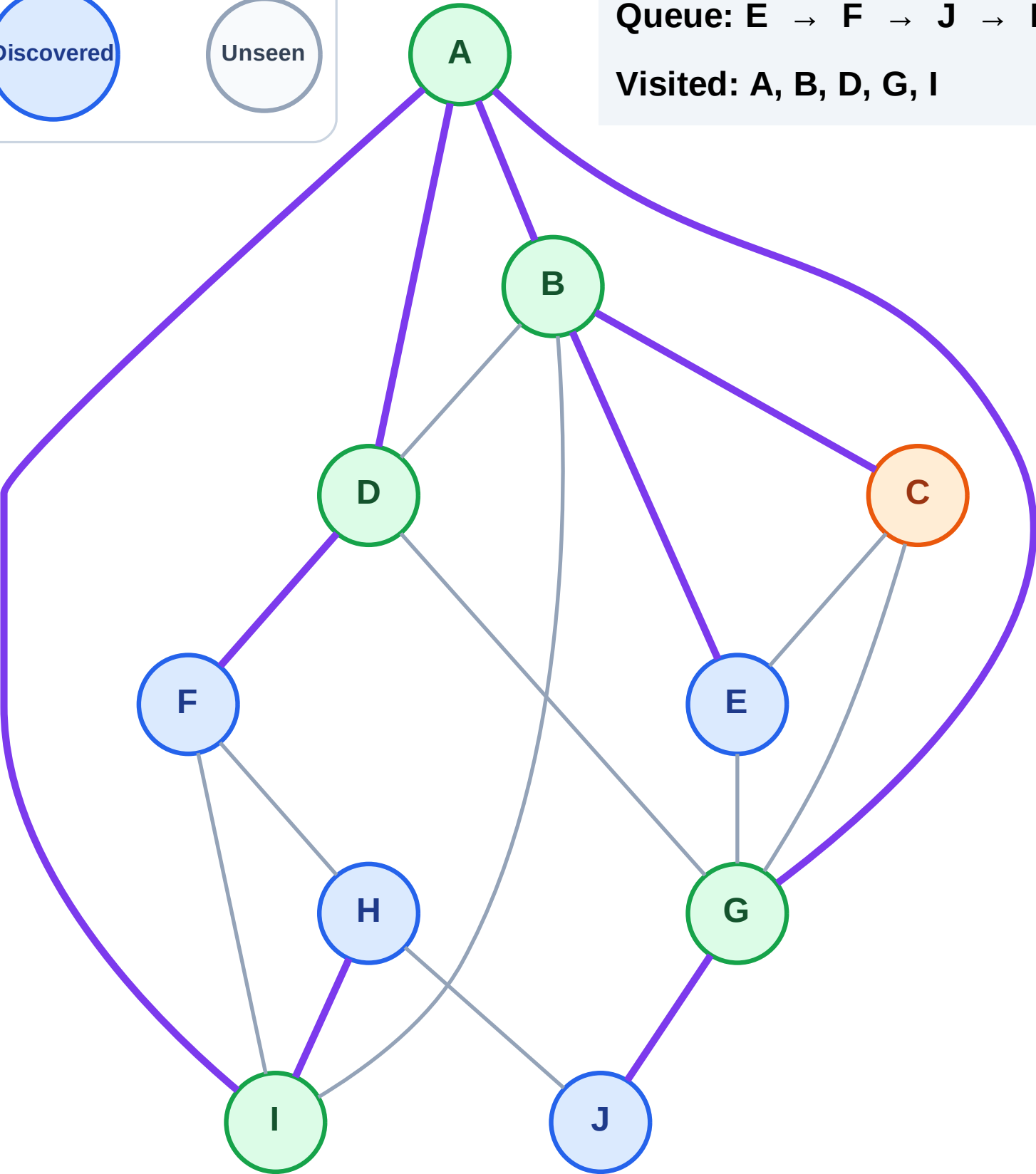
Processing

Discovered

Unseen

Queue: E → F → J → H

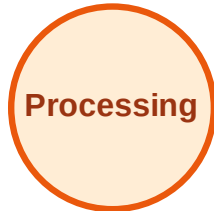
Visited: A, B, D, G, I



BFS Traversal

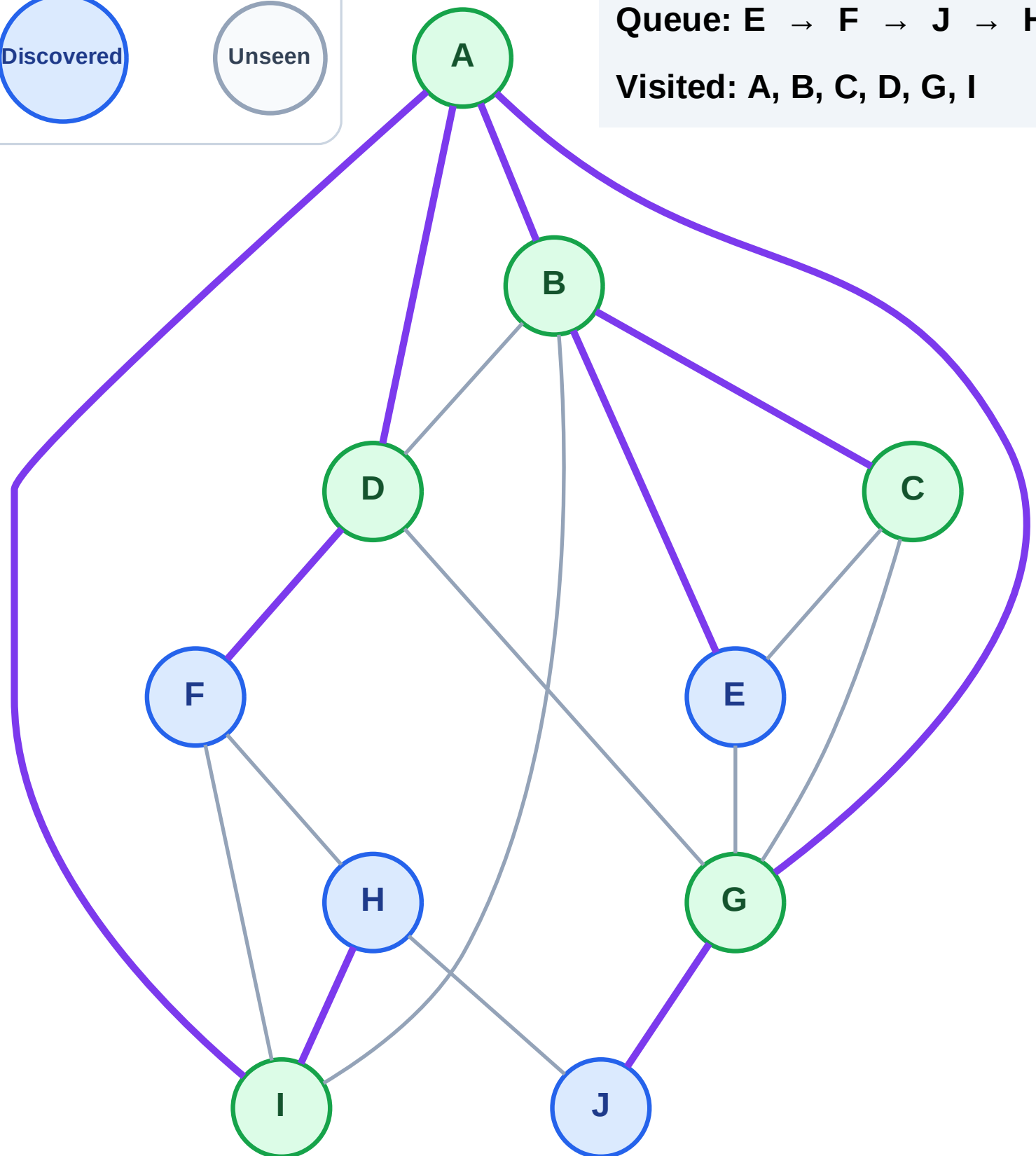
Step 13: No new neighbors from C

Legend



Queue: E → F → J → H

Visited: A, B, C, D, G, I



BFS Traversal

Step 14: Process node E

Legend

Complete

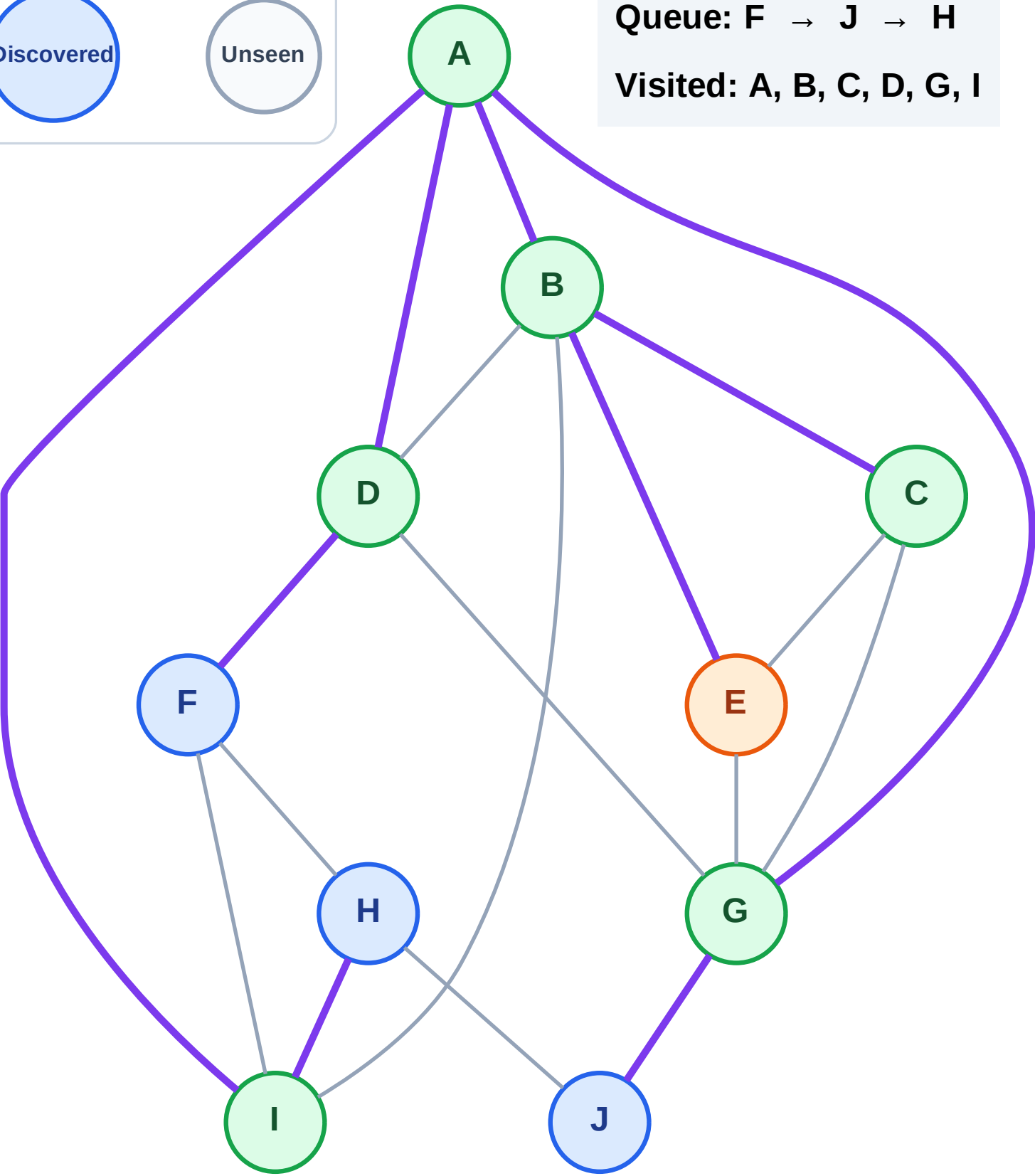
Processing

Discovered

Unseen

Queue: F → J → H

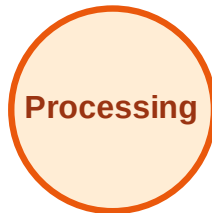
Visited: A, B, C, D, G, I



BFS Traversal

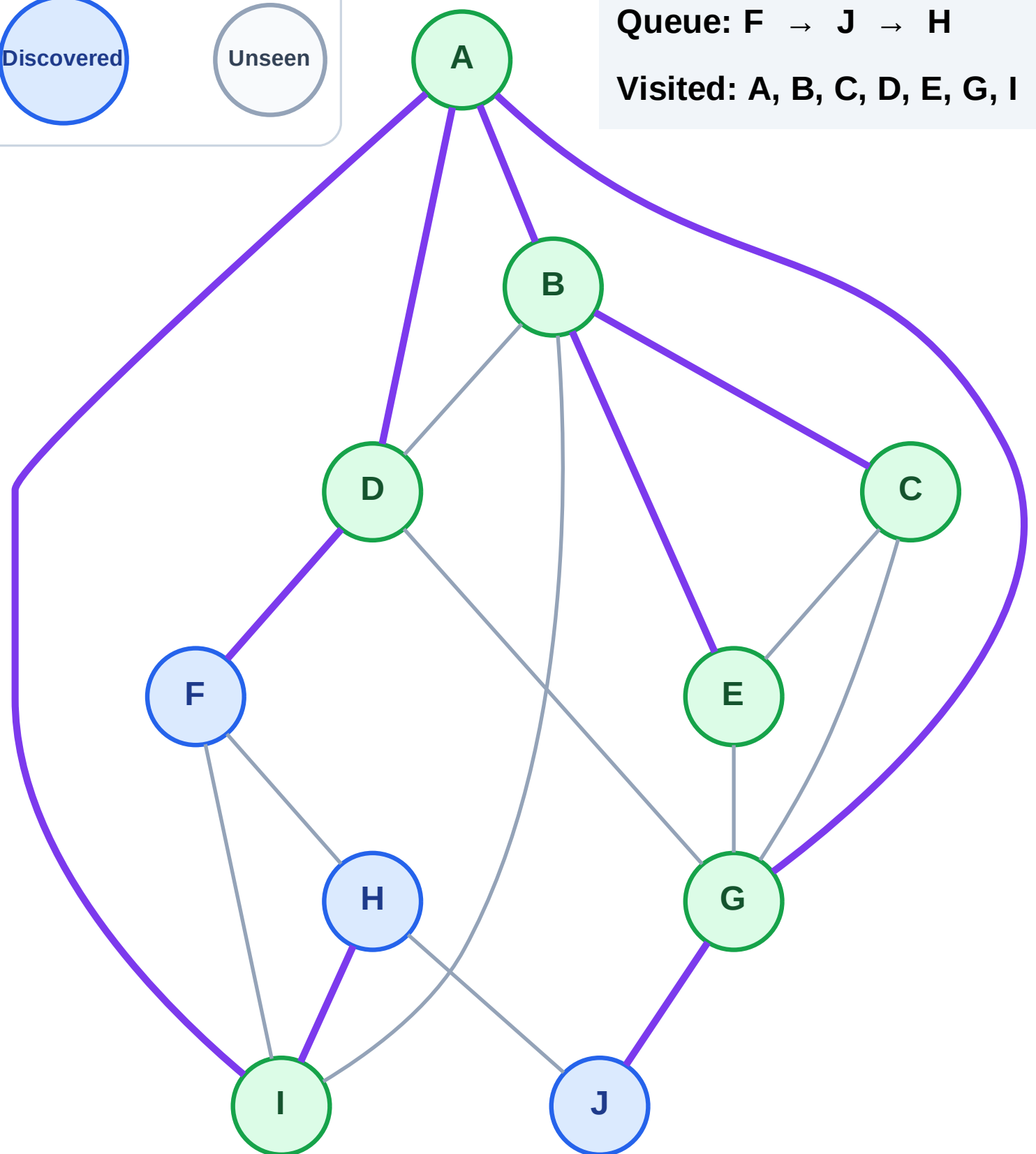
Step 15: No new neighbors from E

Legend



Queue: F → J → H

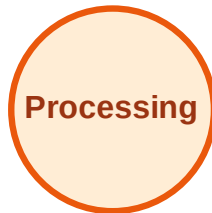
Visited: A, B, C, D, E, G, I



BFS Traversal

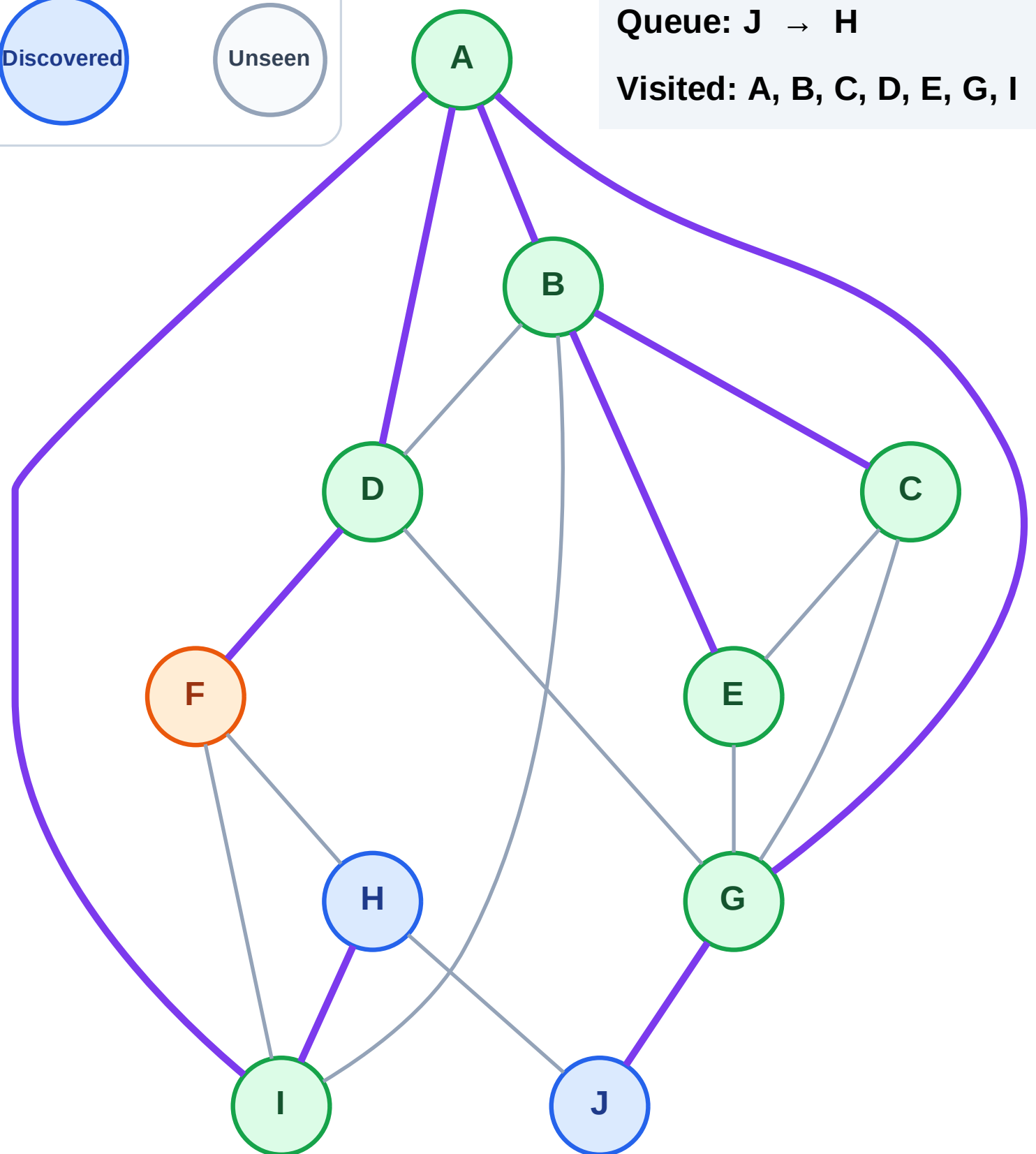
Step 16: Process node F

Legend



Queue: J → H

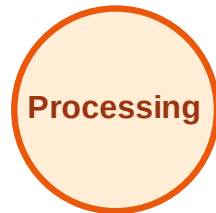
Visited: A, B, C, D, E, G, I



BFS Traversal

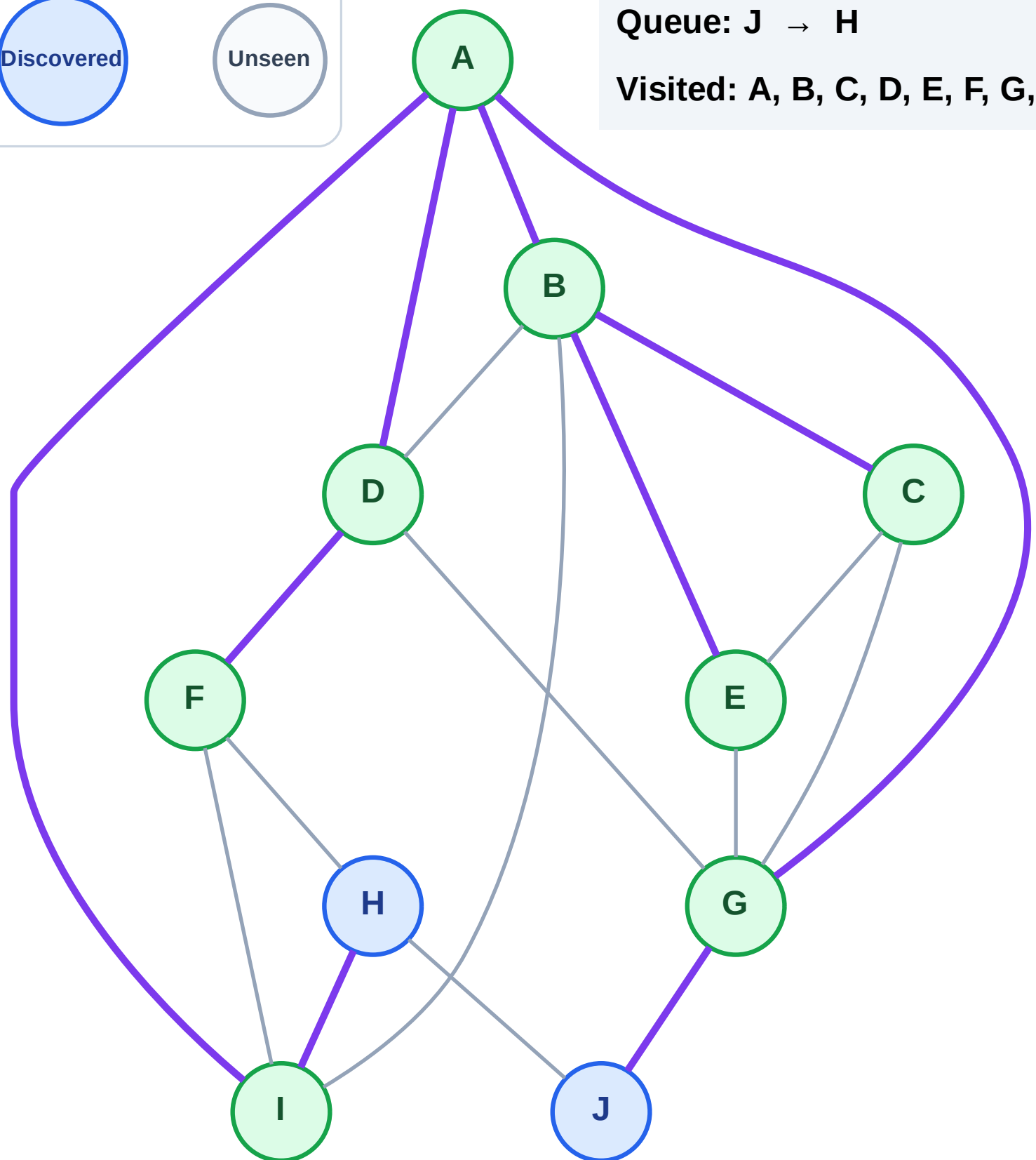
Step 17: No new neighbors from F

Legend



Queue: J → H

Visited: A, B, C, D, E, F, G, I



BFS Traversal

Step 18: Process node J

Legend

Complete

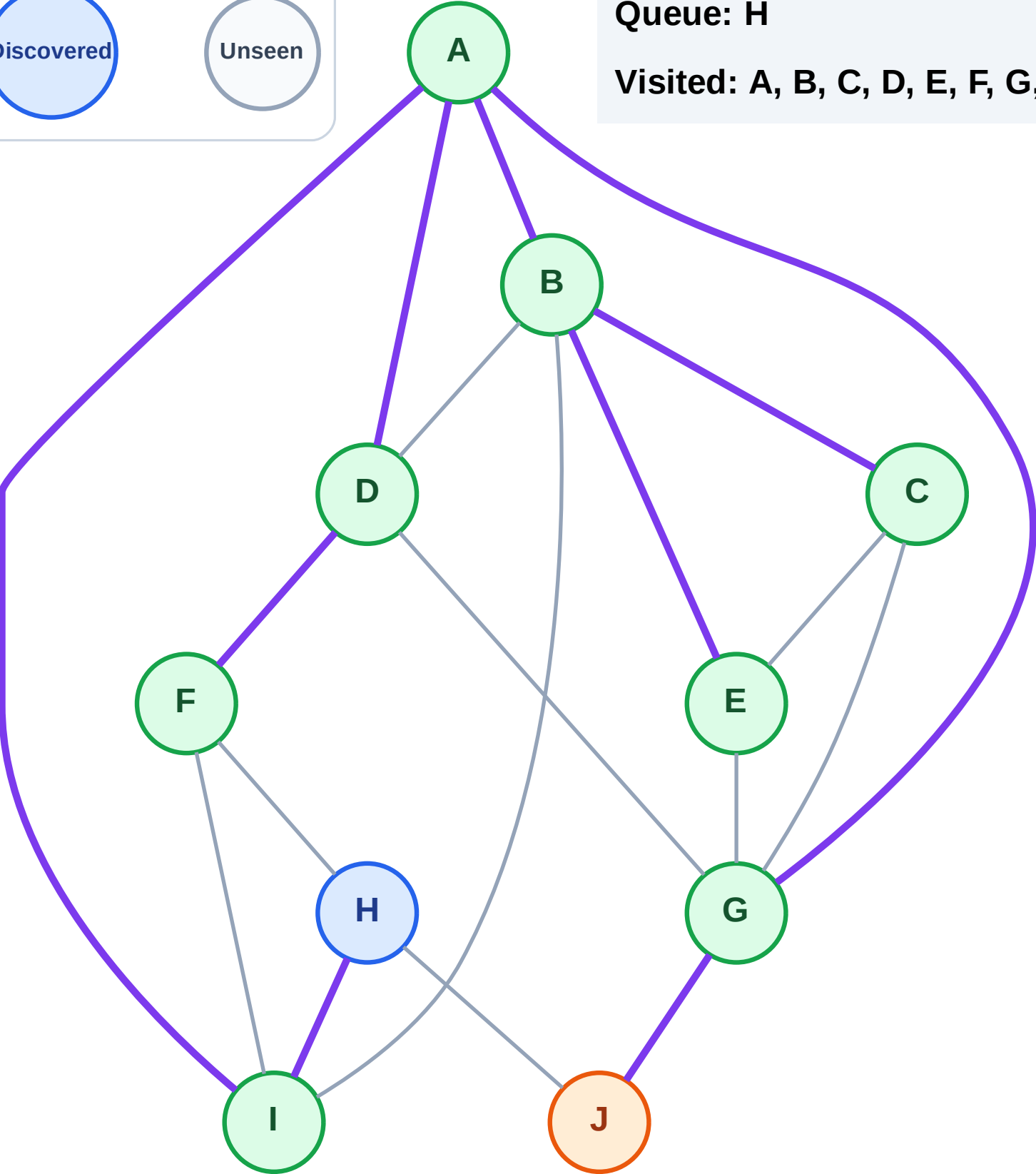
Processing

Discovered

Unseen

Queue: H

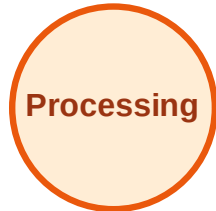
Visited: A, B, C, D, E, F, G, I



BFS Traversal

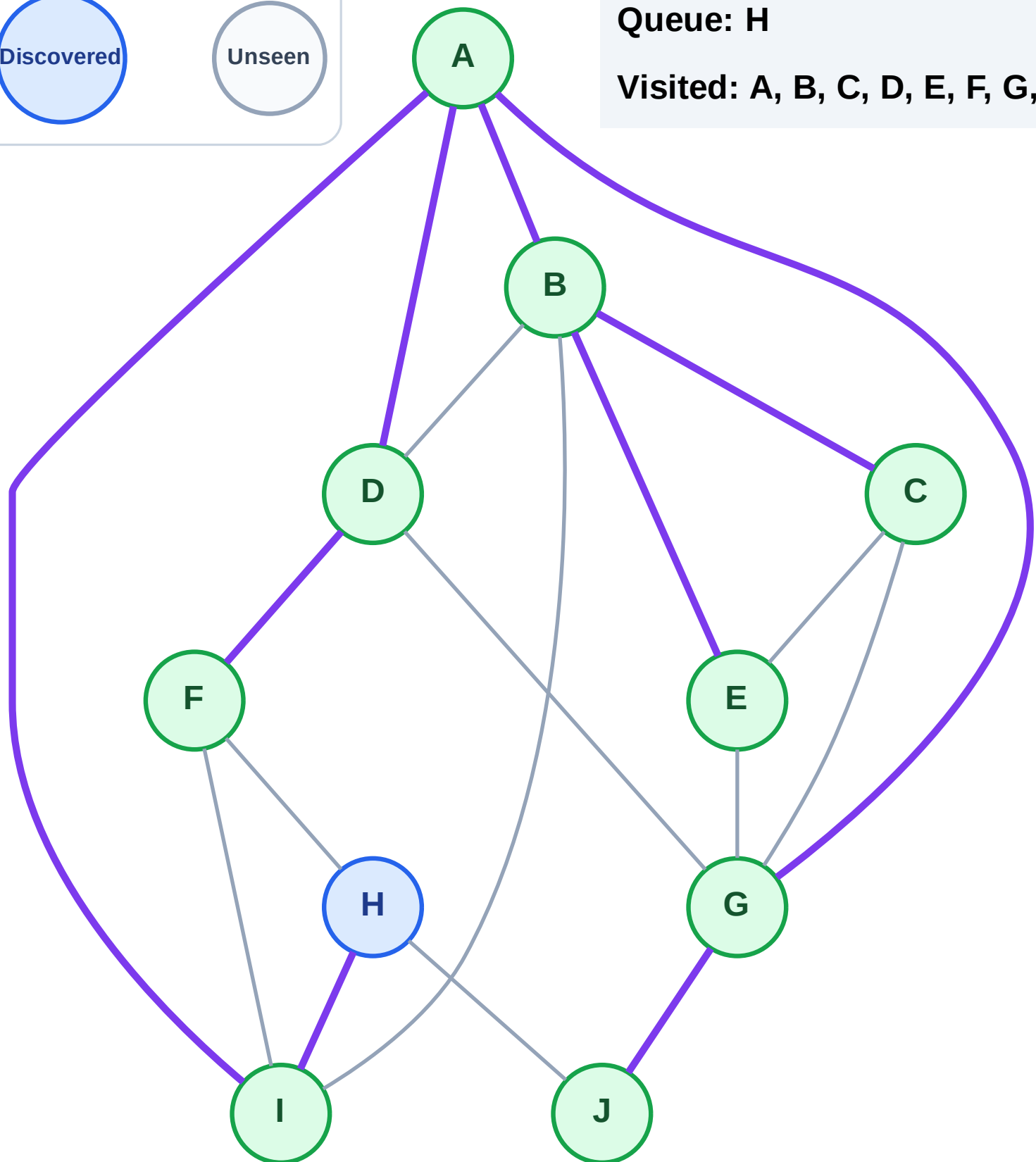
Step 19: No new neighbors from J

Legend



Queue: H

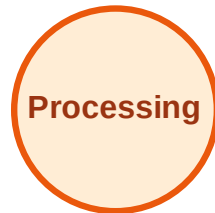
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

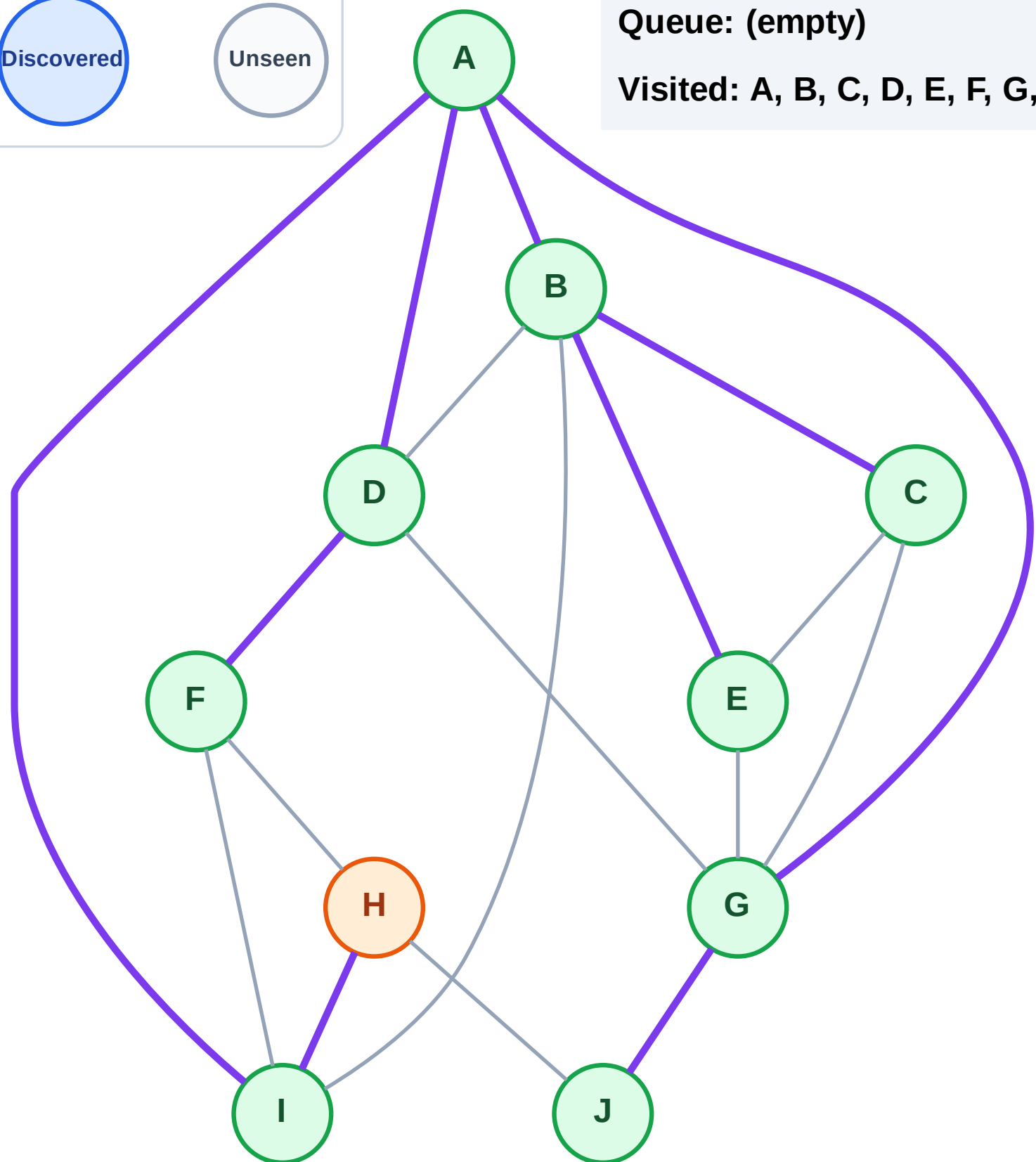
Step 20: Process node H

Legend



Queue: (empty)

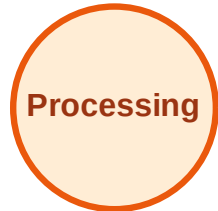
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

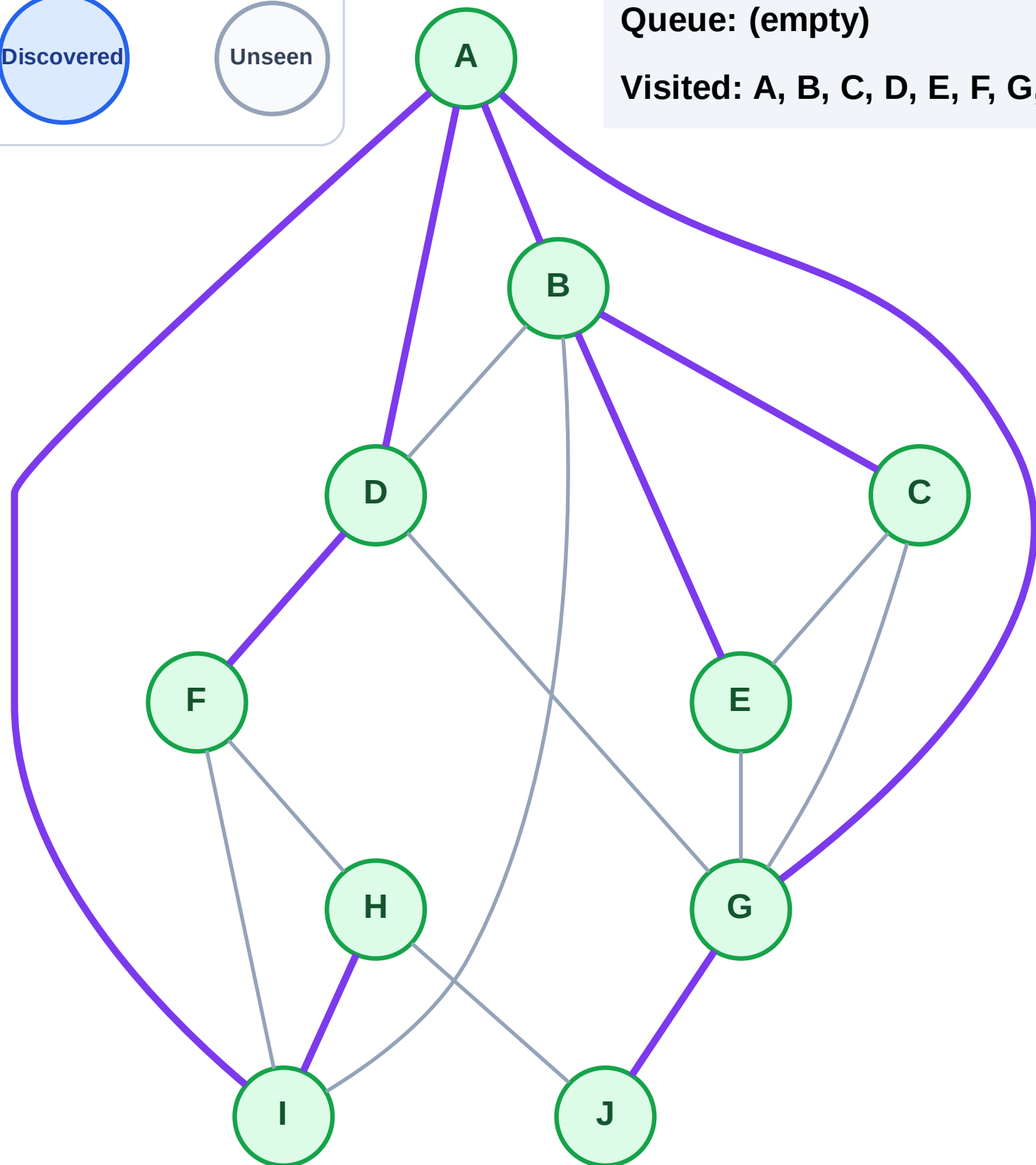
Step 21: No new neighbors from H

Legend



Queue: (empty)

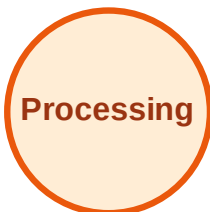
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

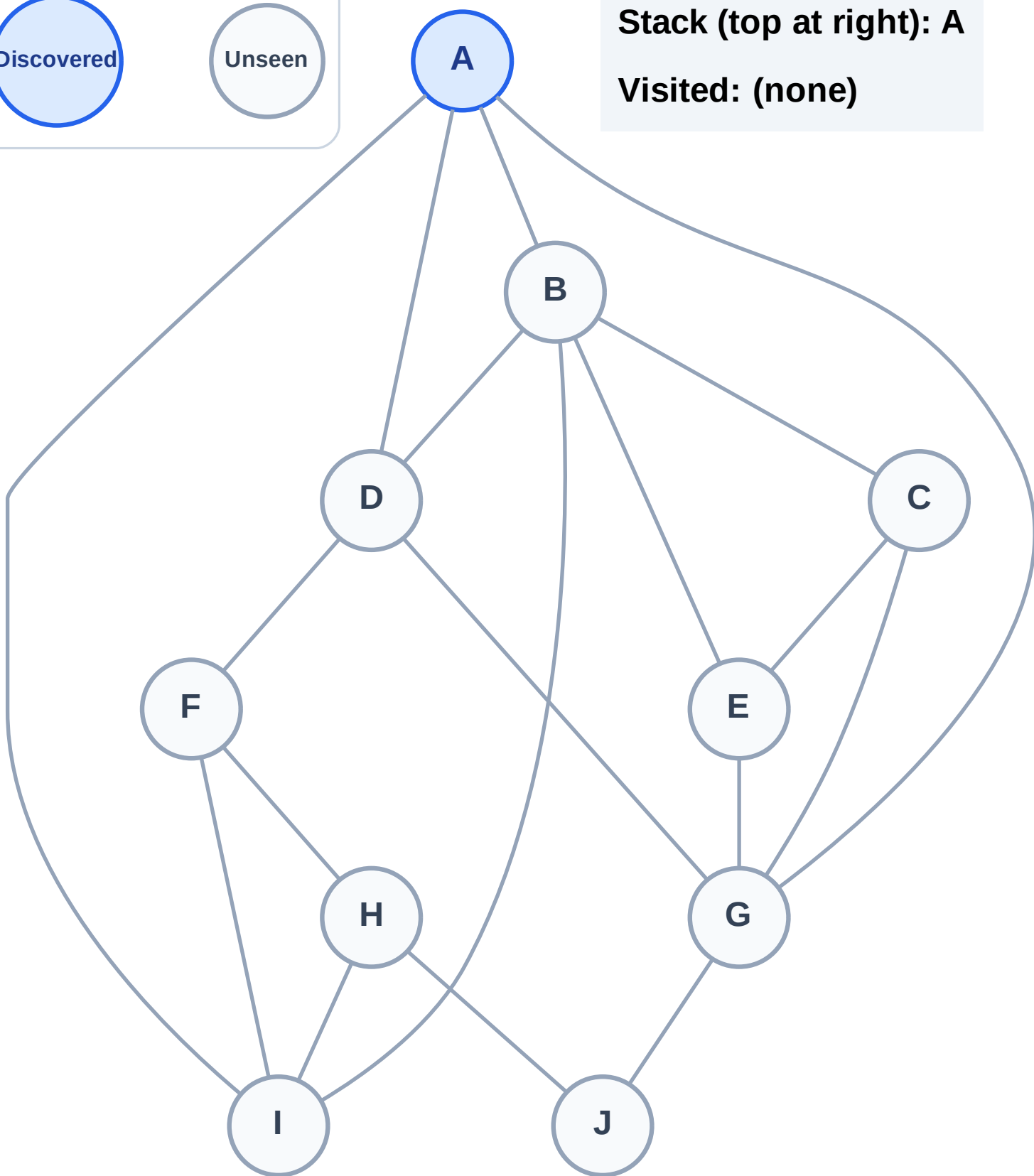
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

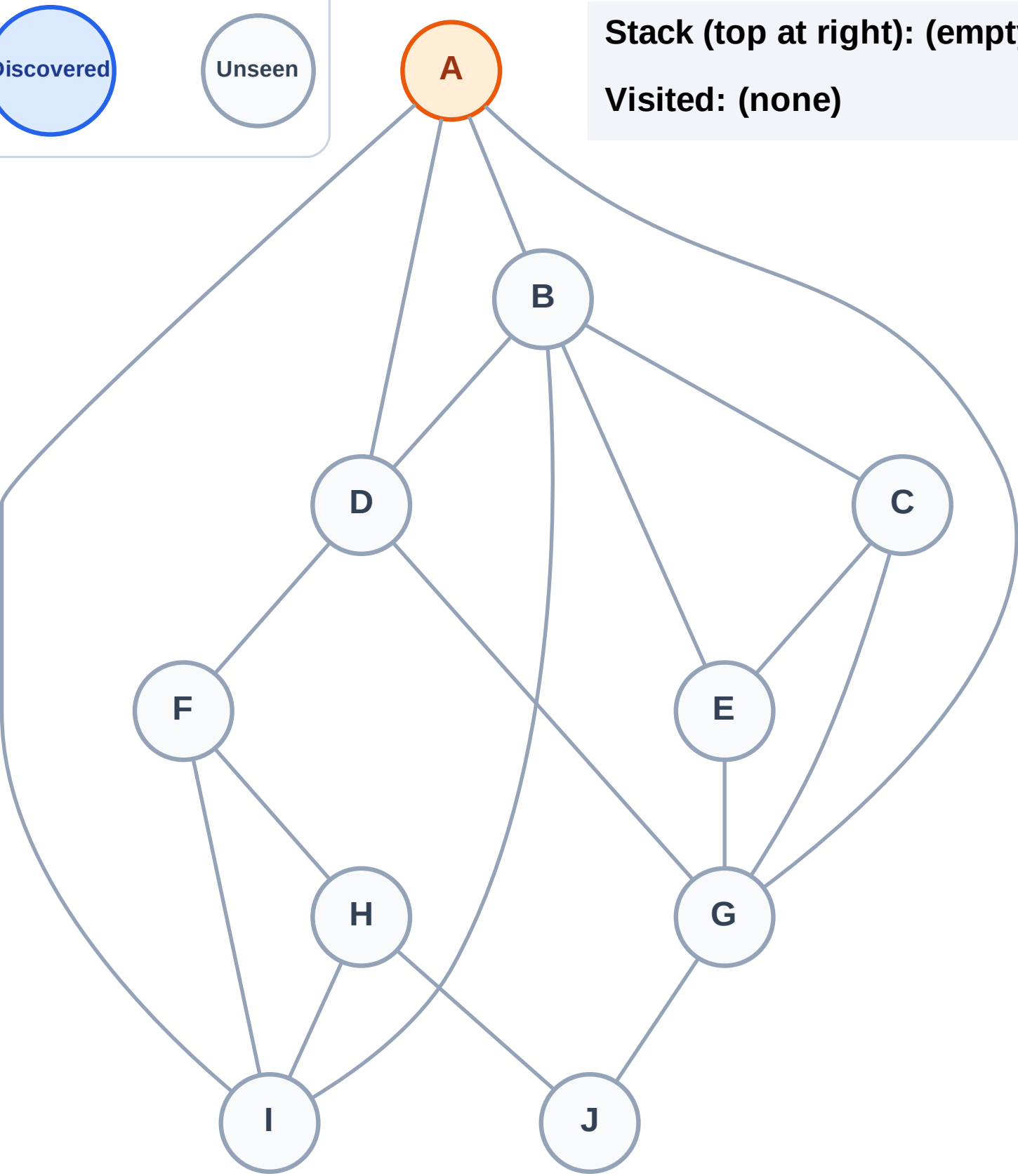
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover I, G, D, B from A

Legend

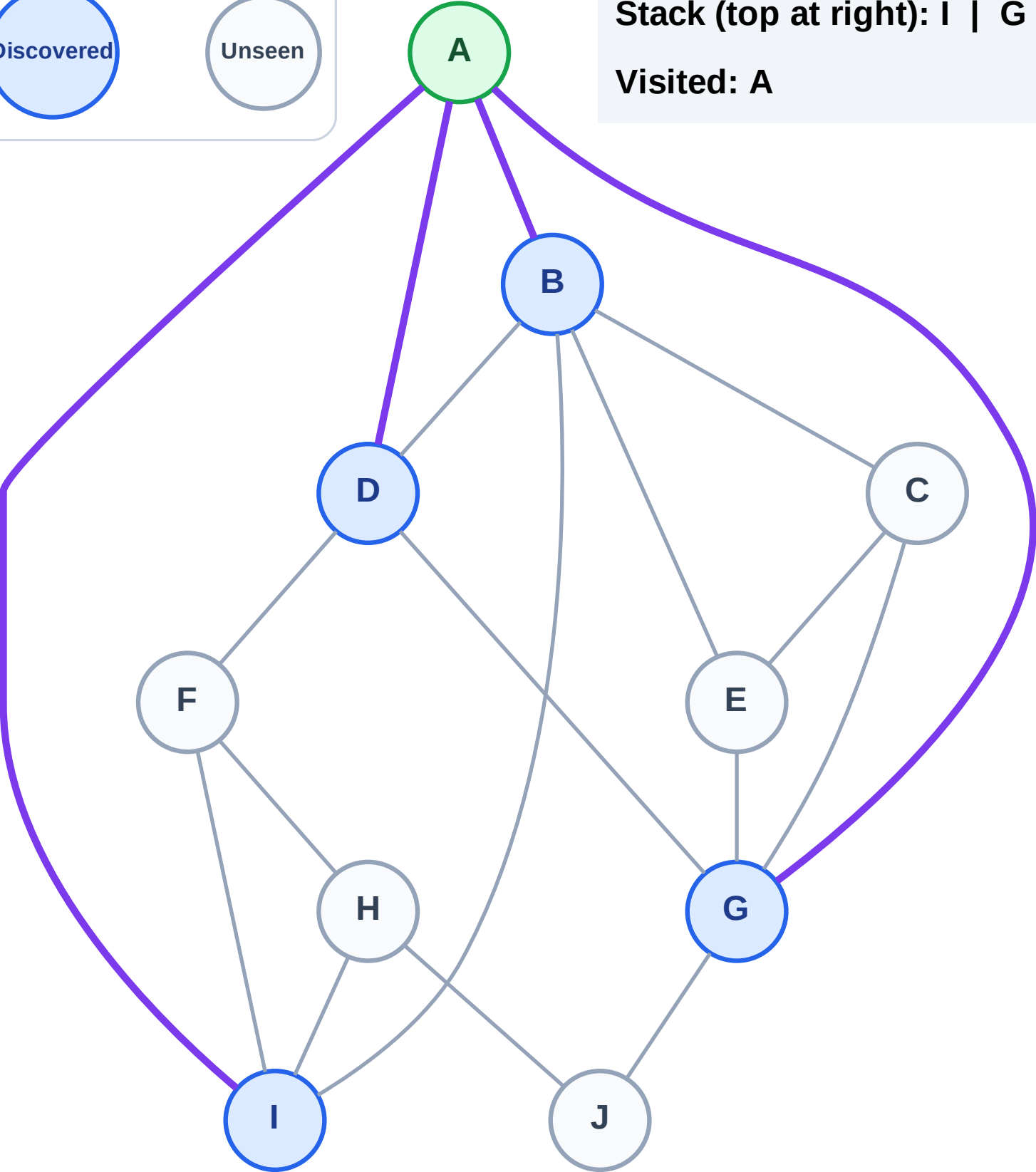
Complete

Processing

Discovered

Unseen

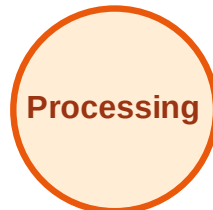
Stack (top at right): I | G | D | B
Visited: A



DFS Traversal

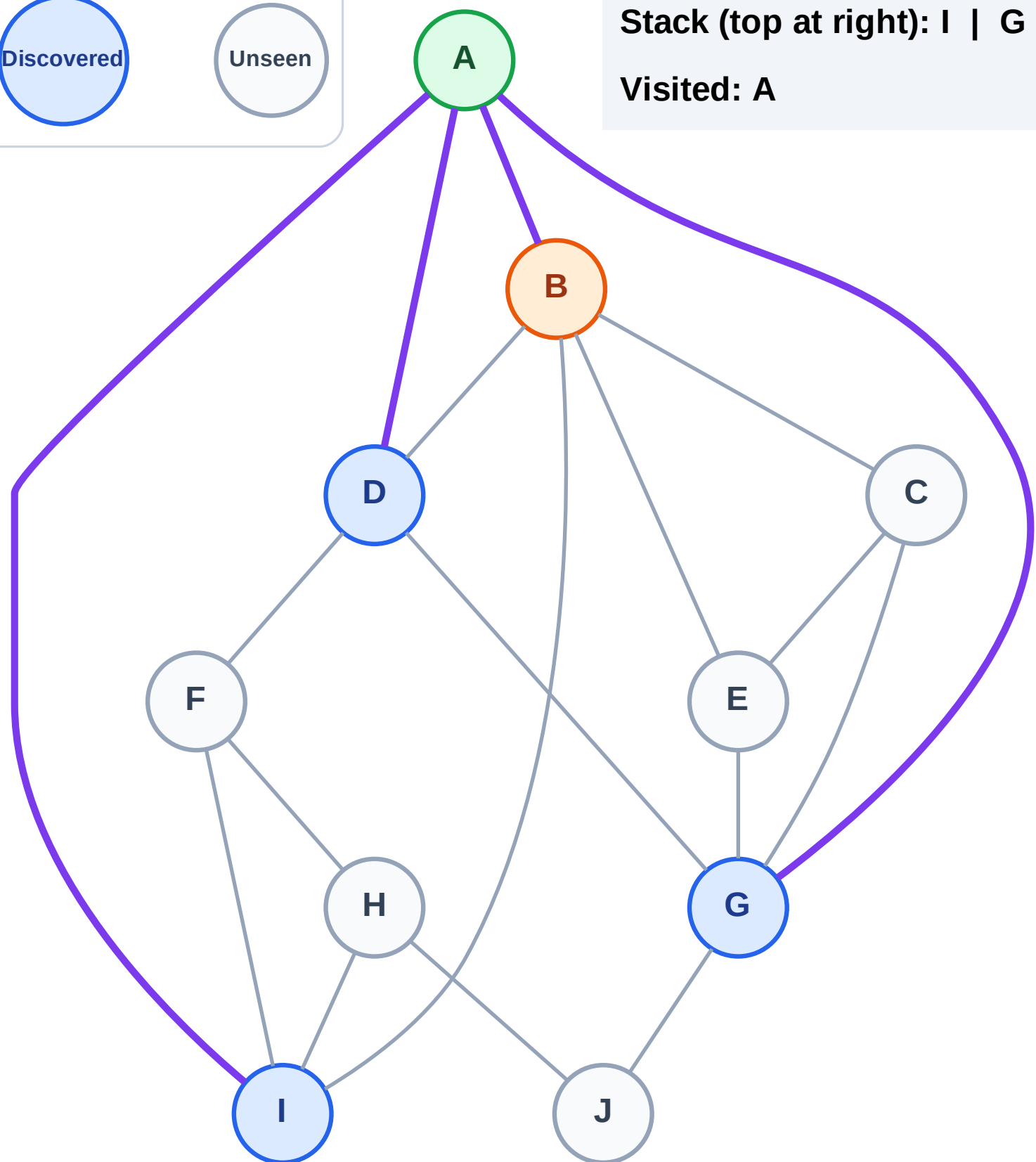
Step 4: Process node B

Legend



Stack (top at right): I | G | D

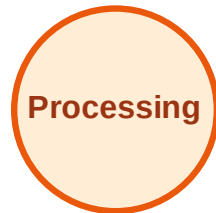
Visited: A



DFS Traversal

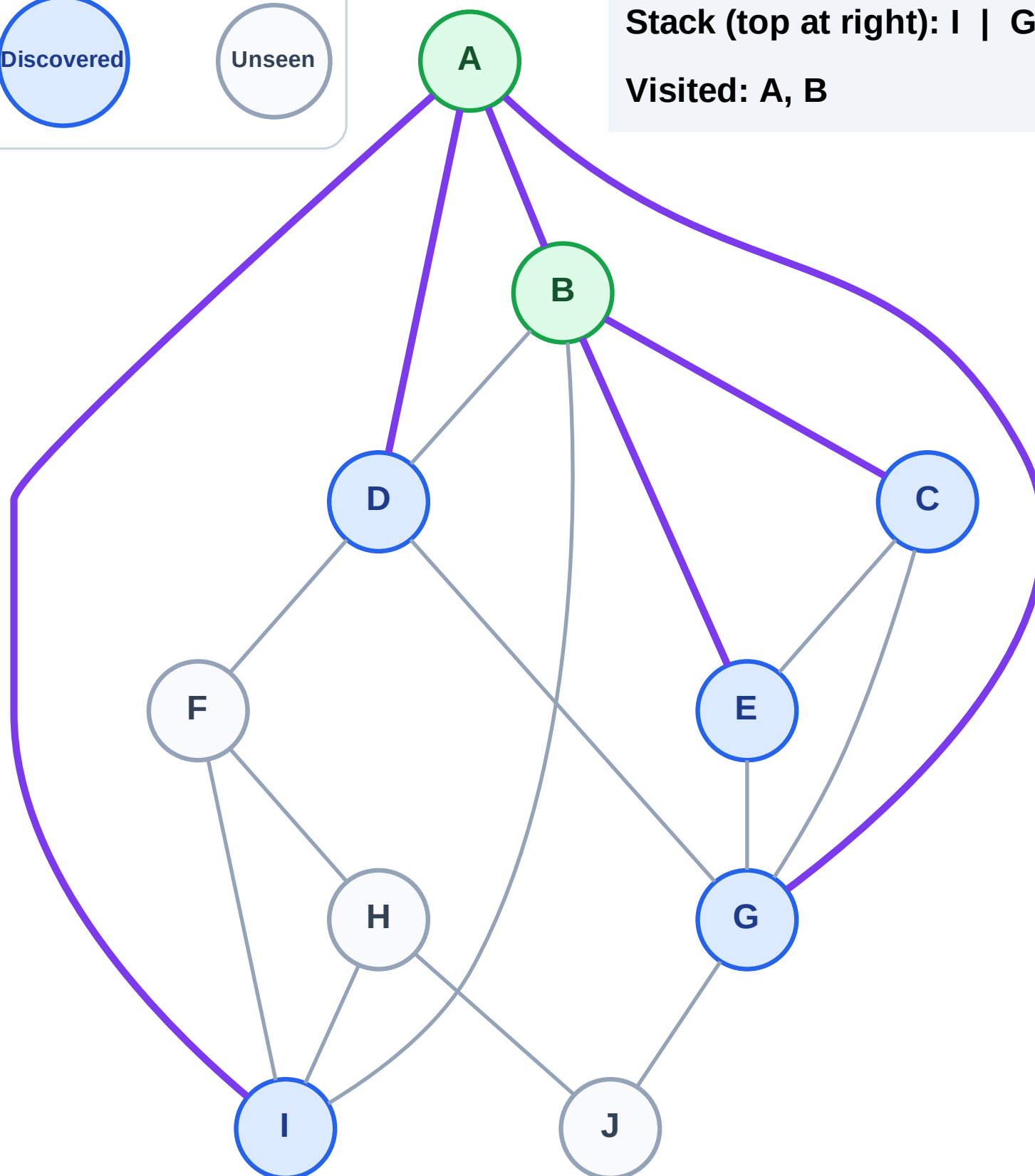
Step 5: Discover E, C from B

Legend



Stack (top at right): I | G | D | E | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

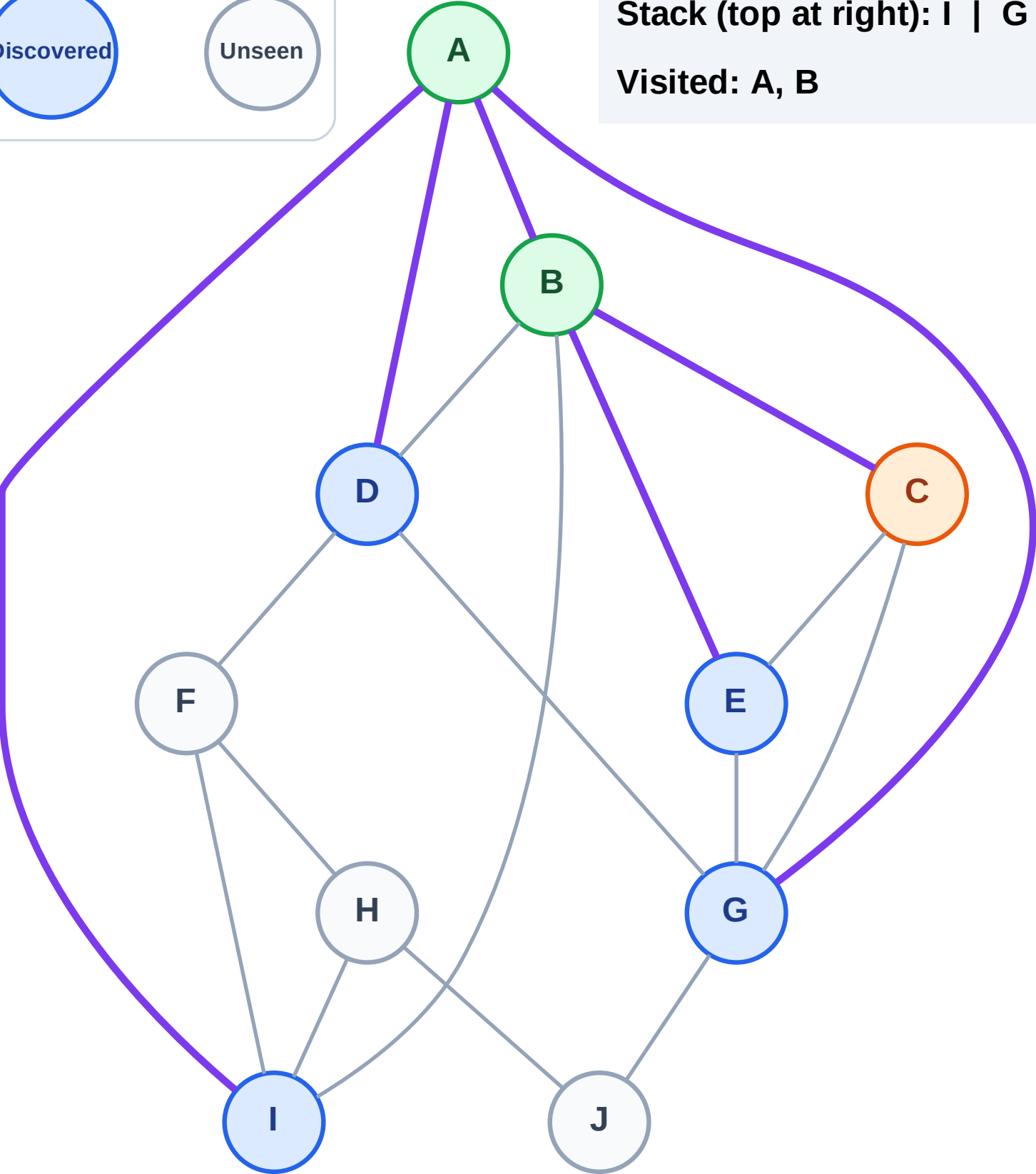
Complete

Processing

Discovered

Unseen

Stack (top at right): I | G | D | E
Visited: A, B



DFS Traversal

Step 7: No new neighbors from C

Legend

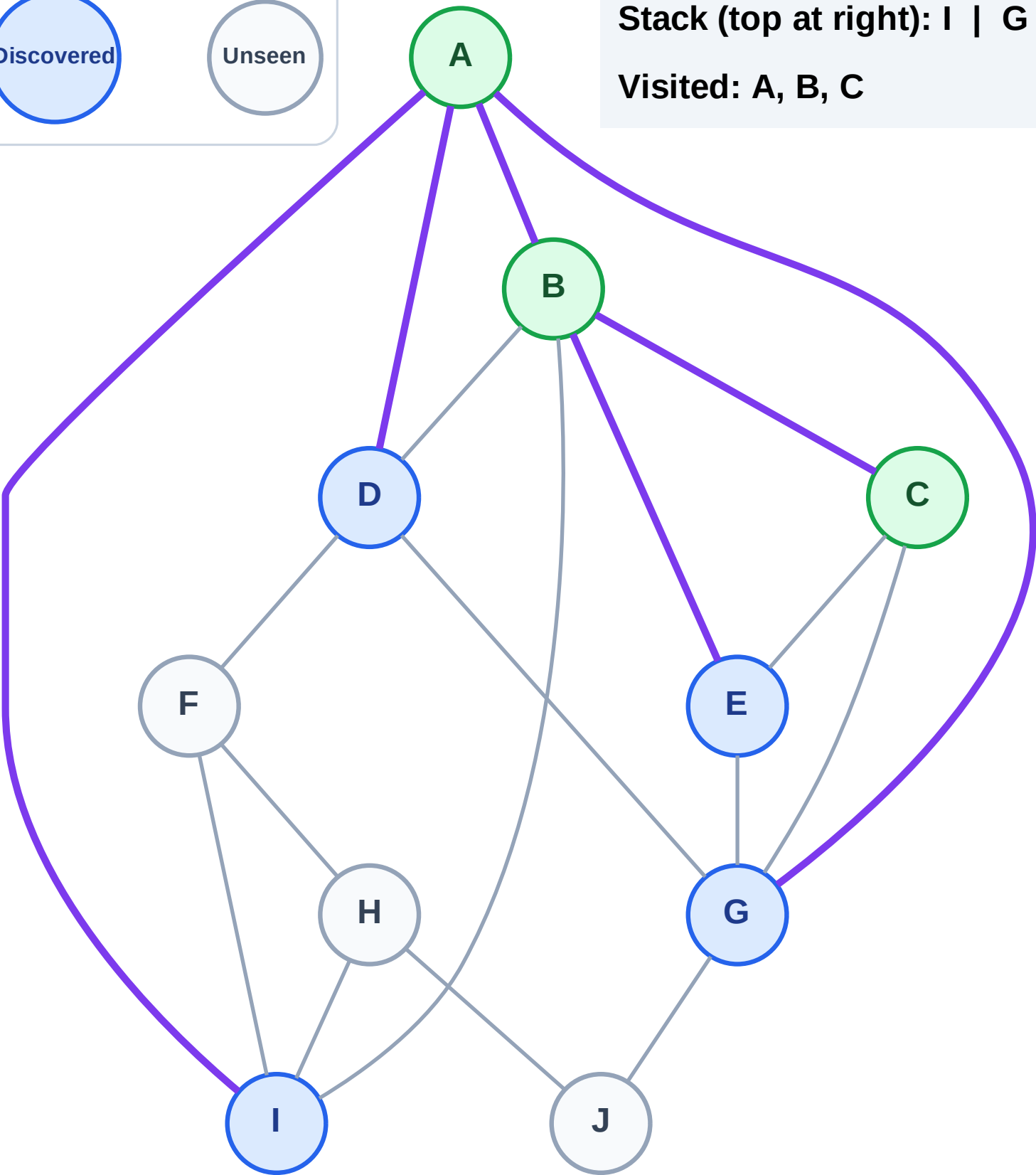
Complete

Processing

Discovered

Unseen

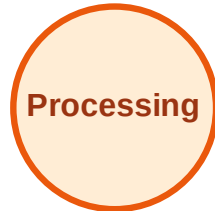
Stack (top at right): I | G | D | E
Visited: A, B, C



DFS Traversal

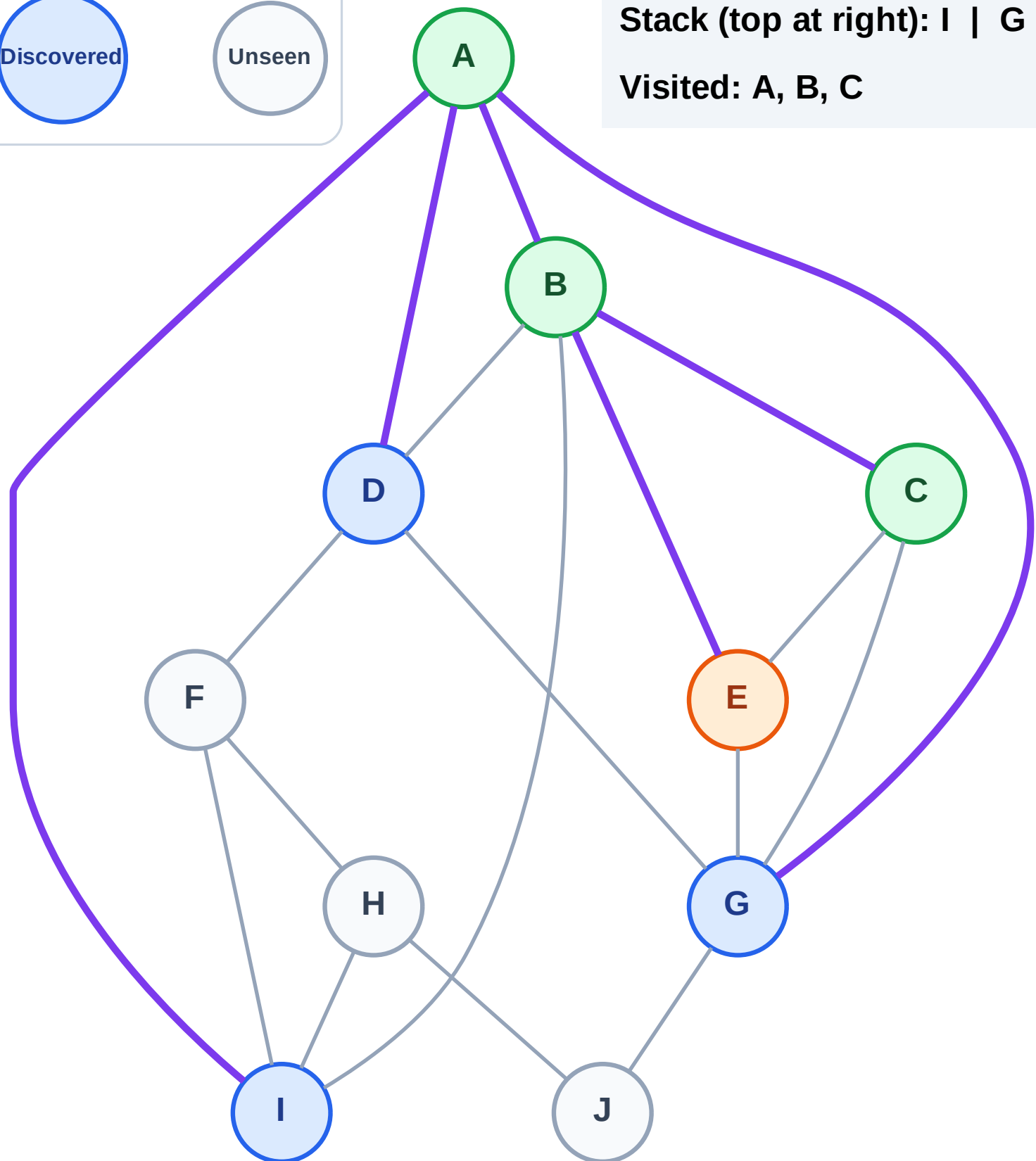
Step 8: Process node E

Legend



Stack (top at right): I | G | D

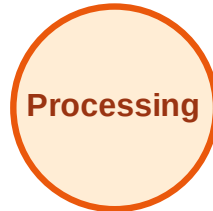
Visited: A, B, C



DFS Traversal

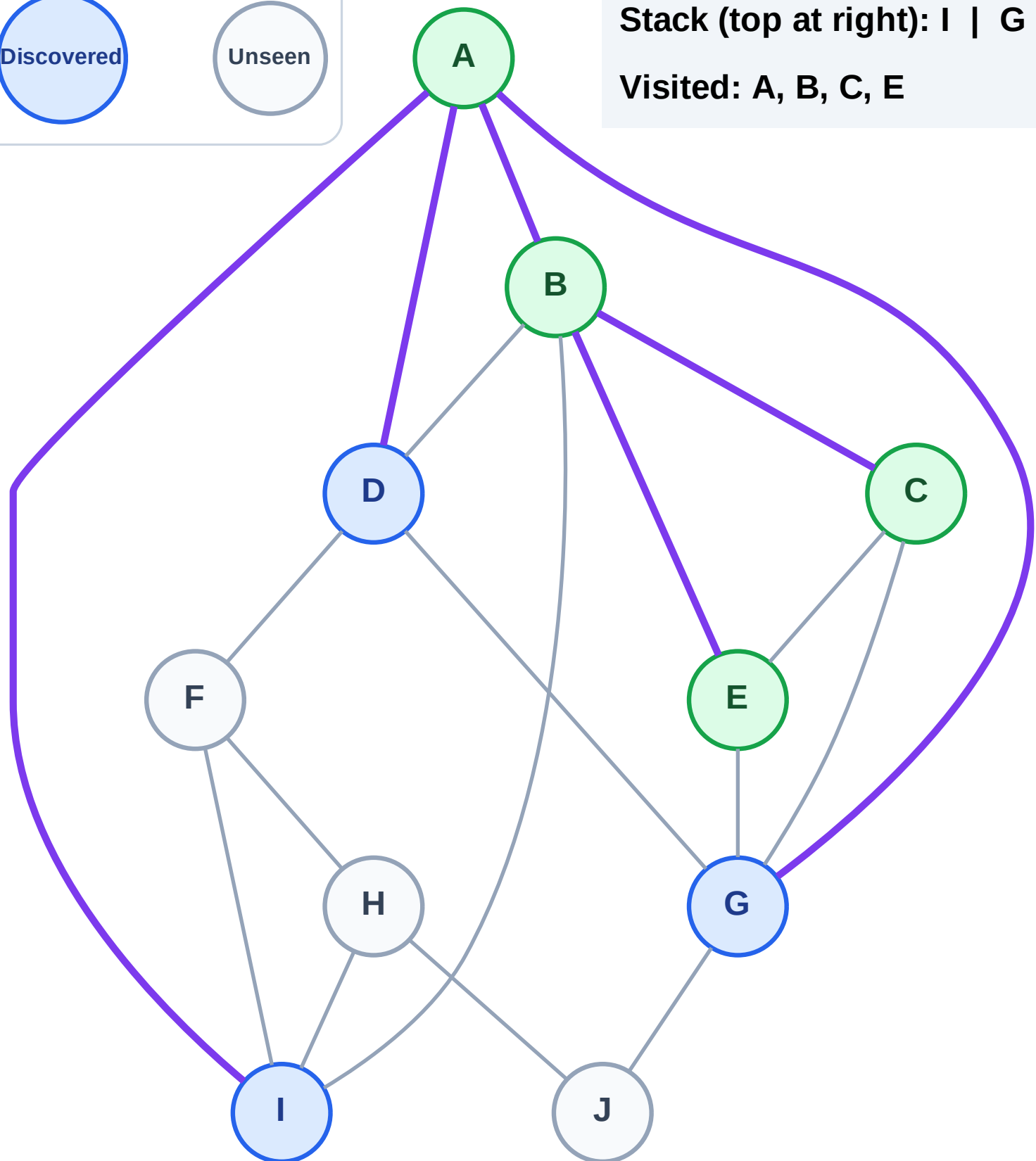
Step 9: No new neighbors from E

Legend



Stack (top at right): I | G | D

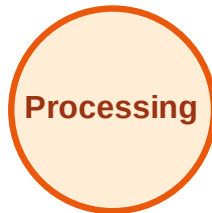
Visited: A, B, C, E



DFS Traversal

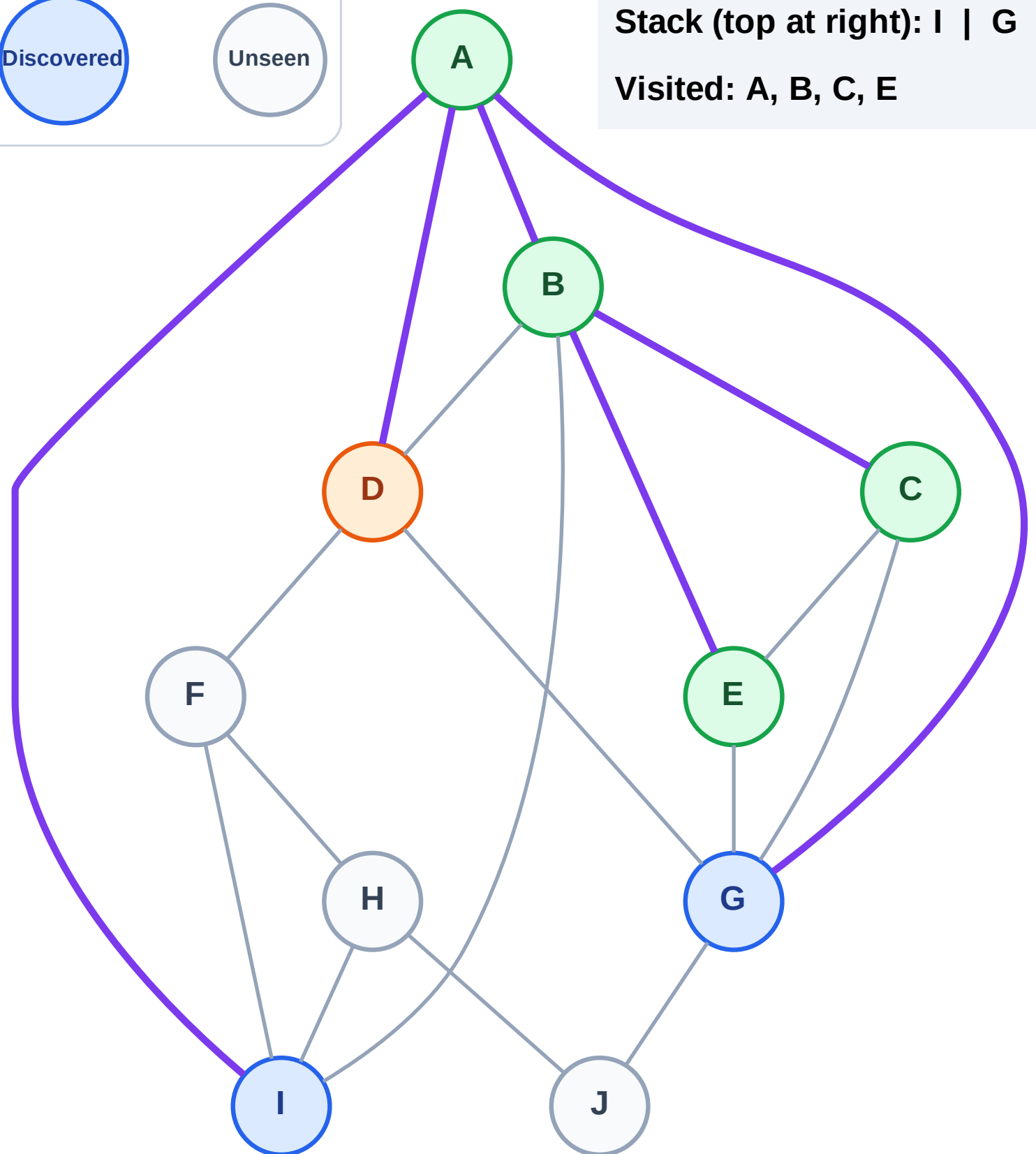
Step 10: Process node D

Legend



Stack (top at right): I | G

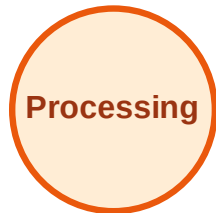
Visited: A, B, C, E



DFS Traversal

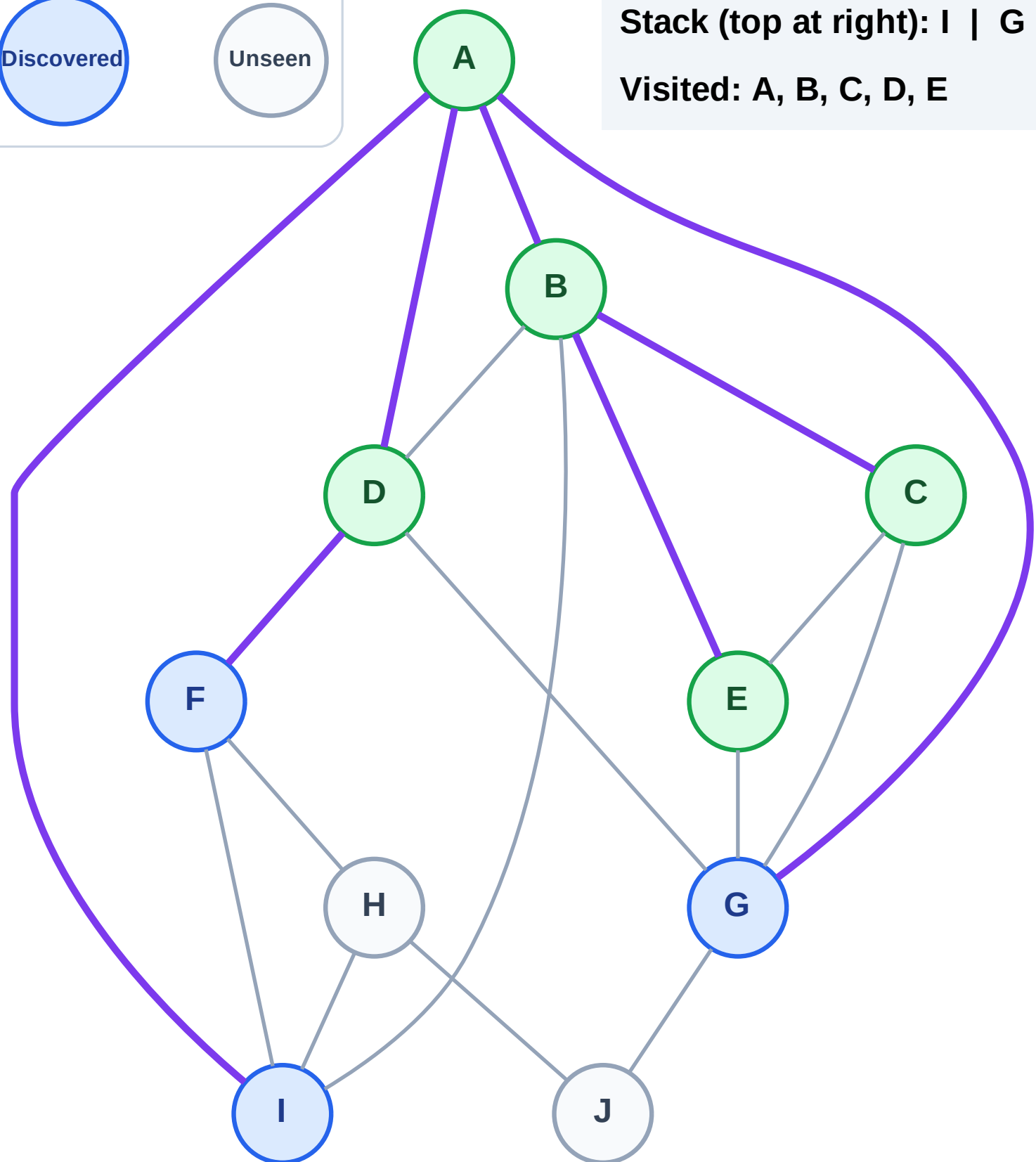
Step 11: Discover F from D

Legend



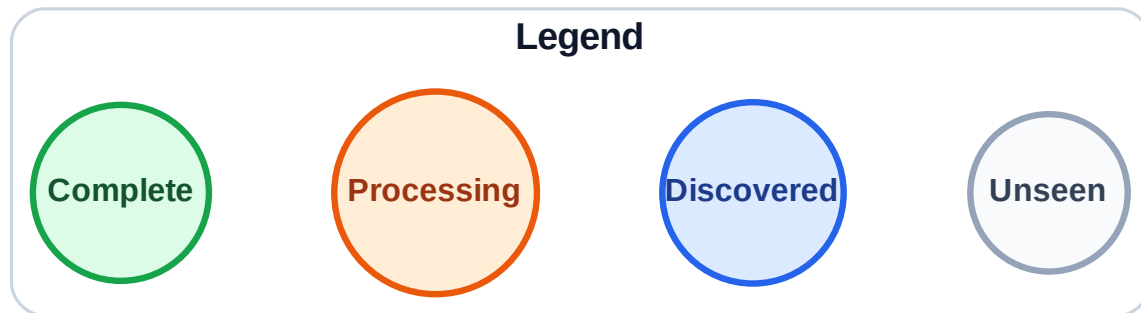
Stack (top at right): I | G | F

Visited: A, B, C, D, E



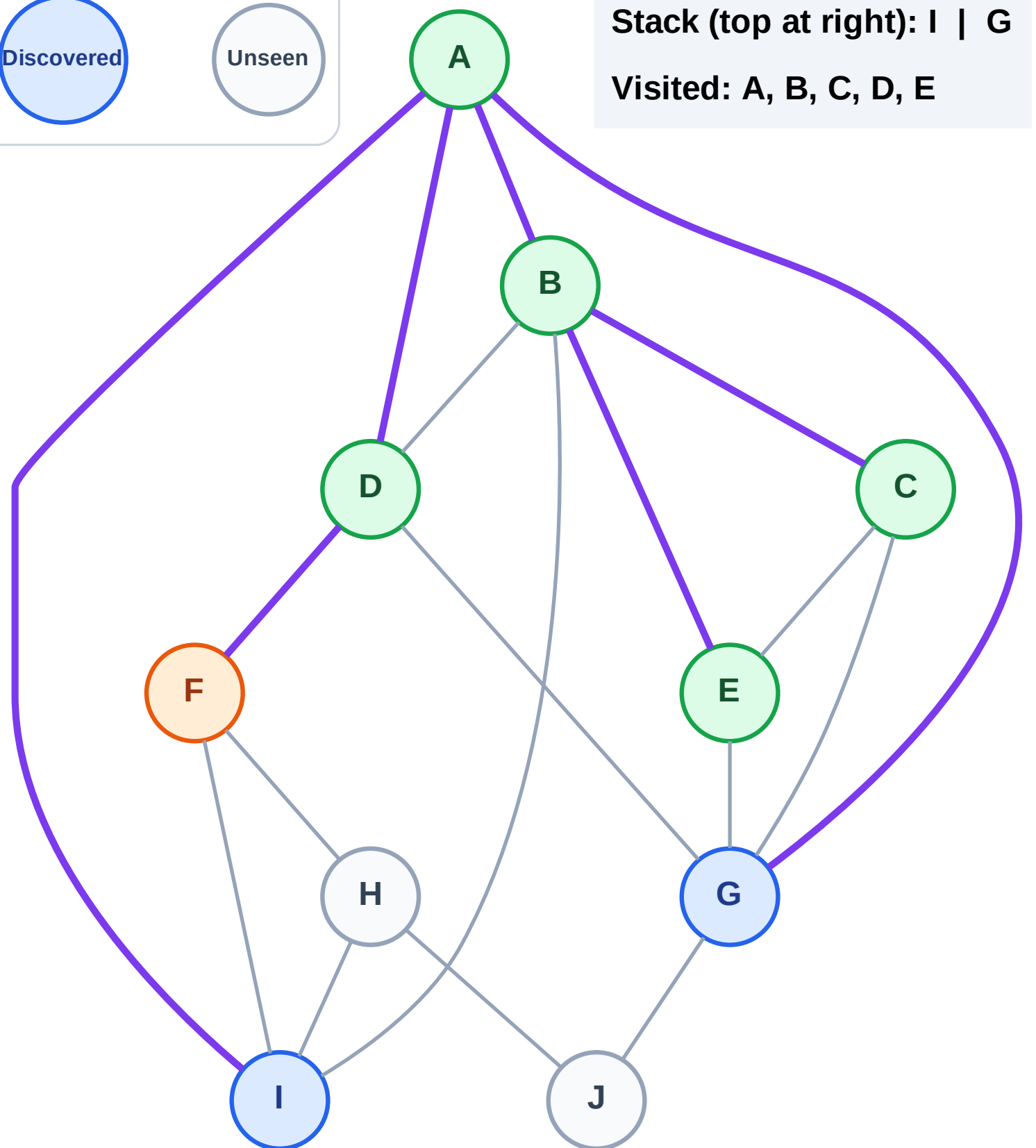
Step 12: Process node F

Step 12: Process node F



Stack (top at right): I | G

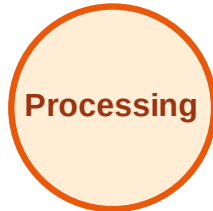
Visited: A, B, C, D, E



DFS Traversal

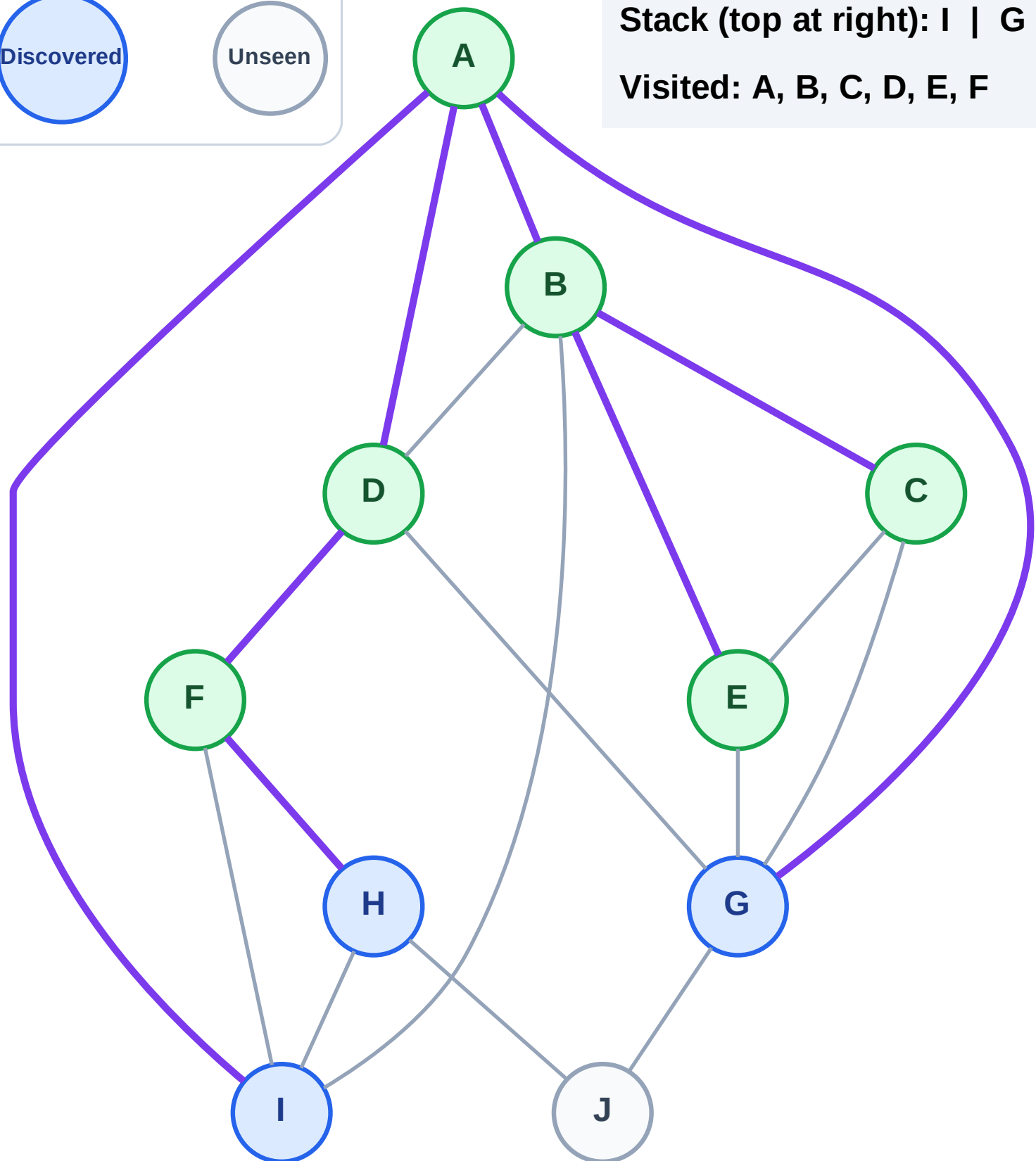
Step 13: Discover H from F

Legend



Stack (top at right): I | G | H

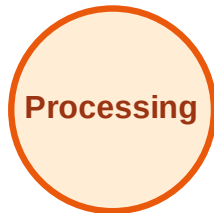
Visited: A, B, C, D, E, F



DFS Traversal

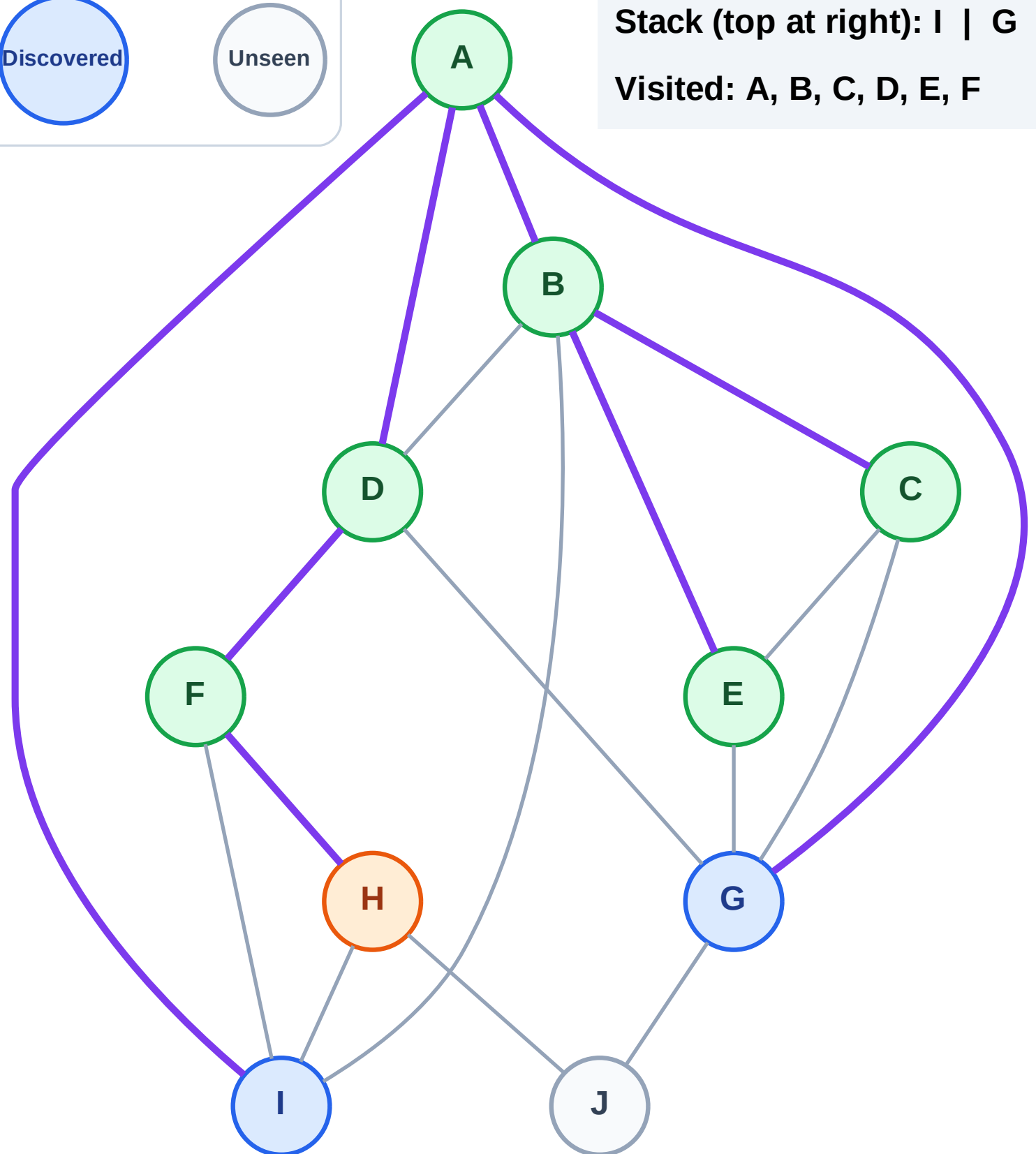
Step 14: Process node H

Legend



Stack (top at right): I | G

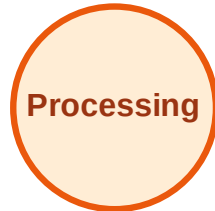
Visited: A, B, C, D, E, F



DFS Traversal

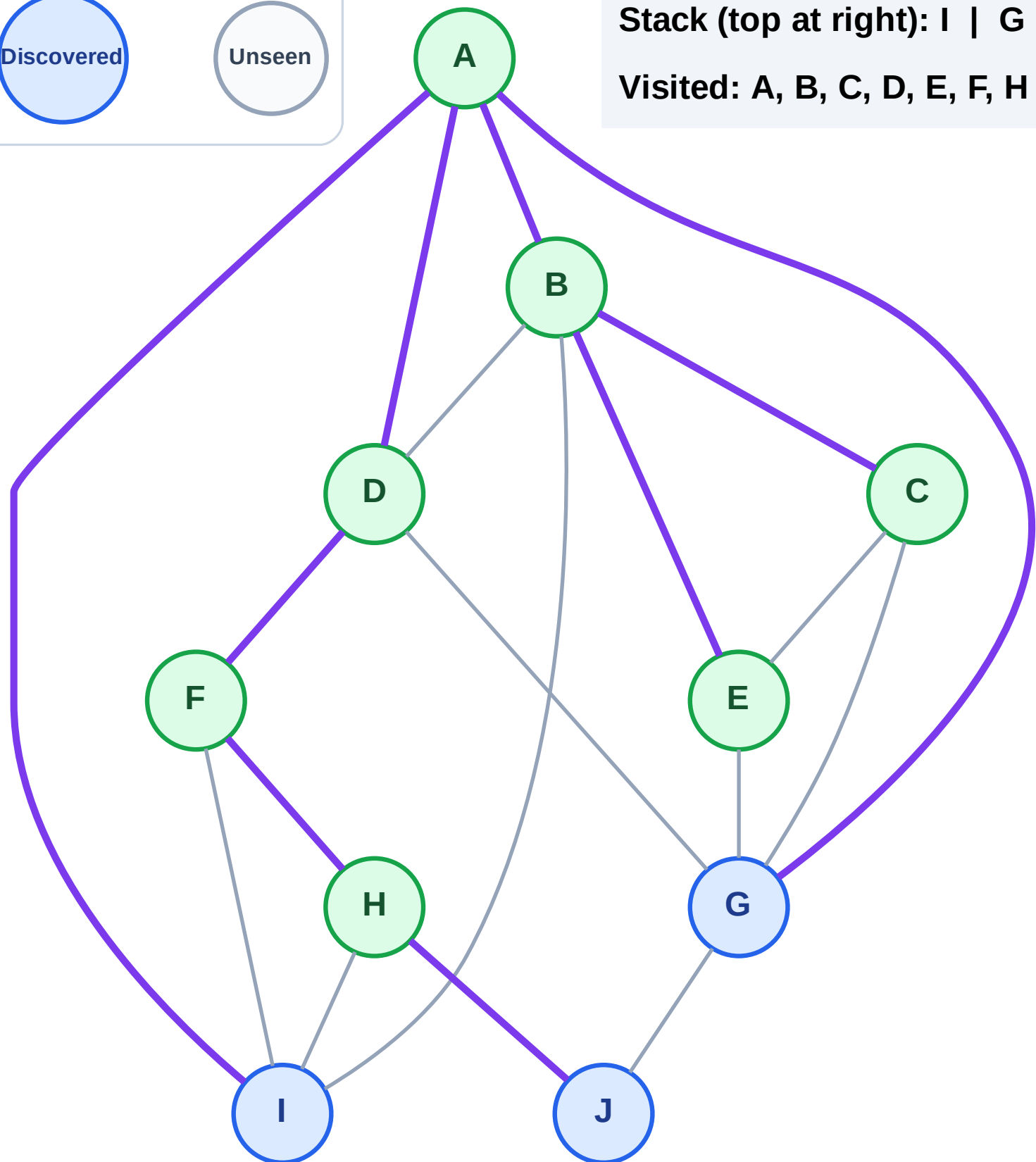
Step 15: Discover J from H

Legend



Stack (top at right): I | G | J

Visited: A, B, C, D, E, F, H



DFS Traversal

Step 16: Process node J

Legend

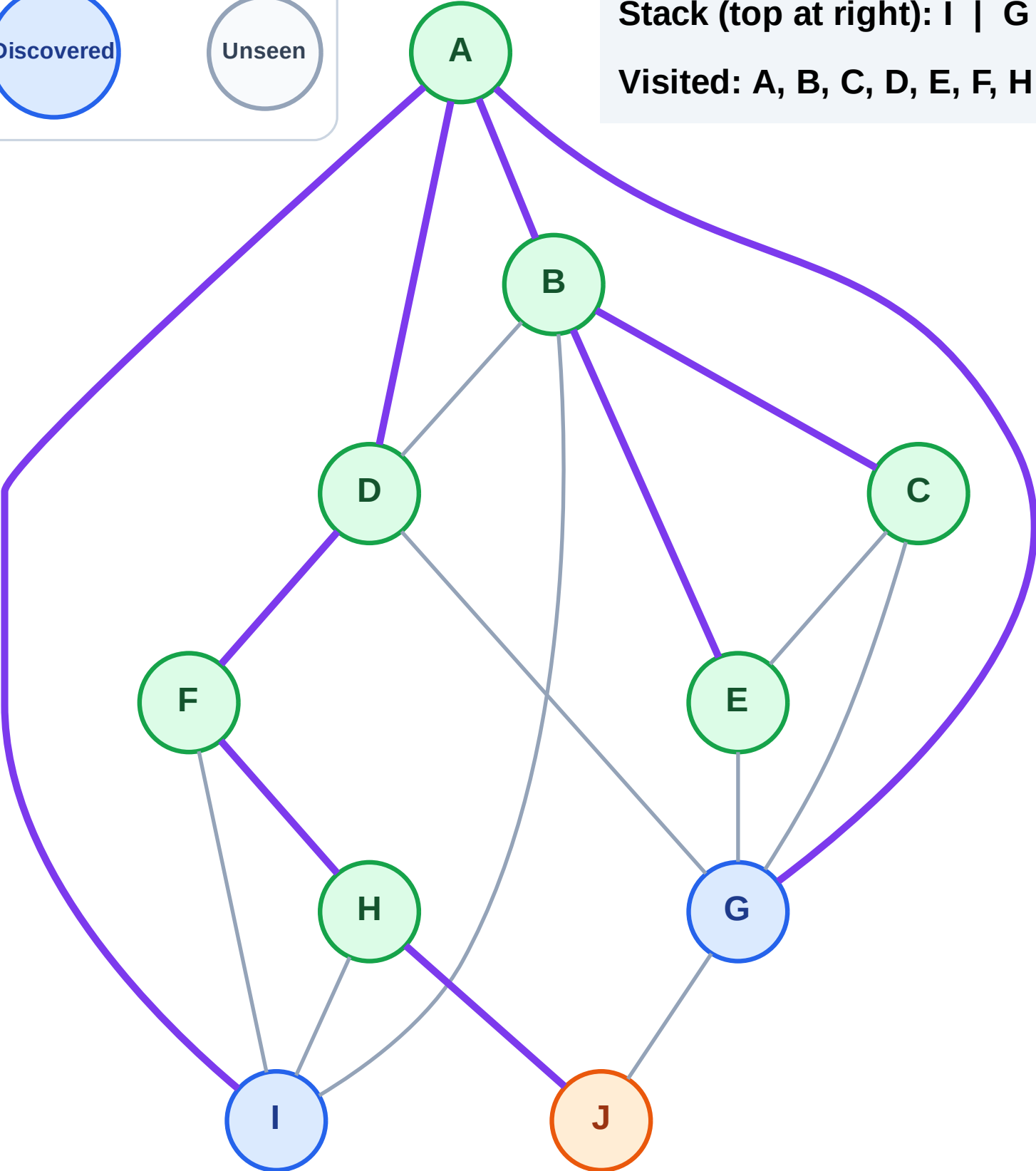
Complete

Processing

Discovered

Unseen

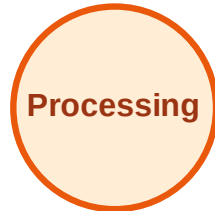
Stack (top at right): I | G
Visited: A, B, C, D, E, F, H



DFS Traversal

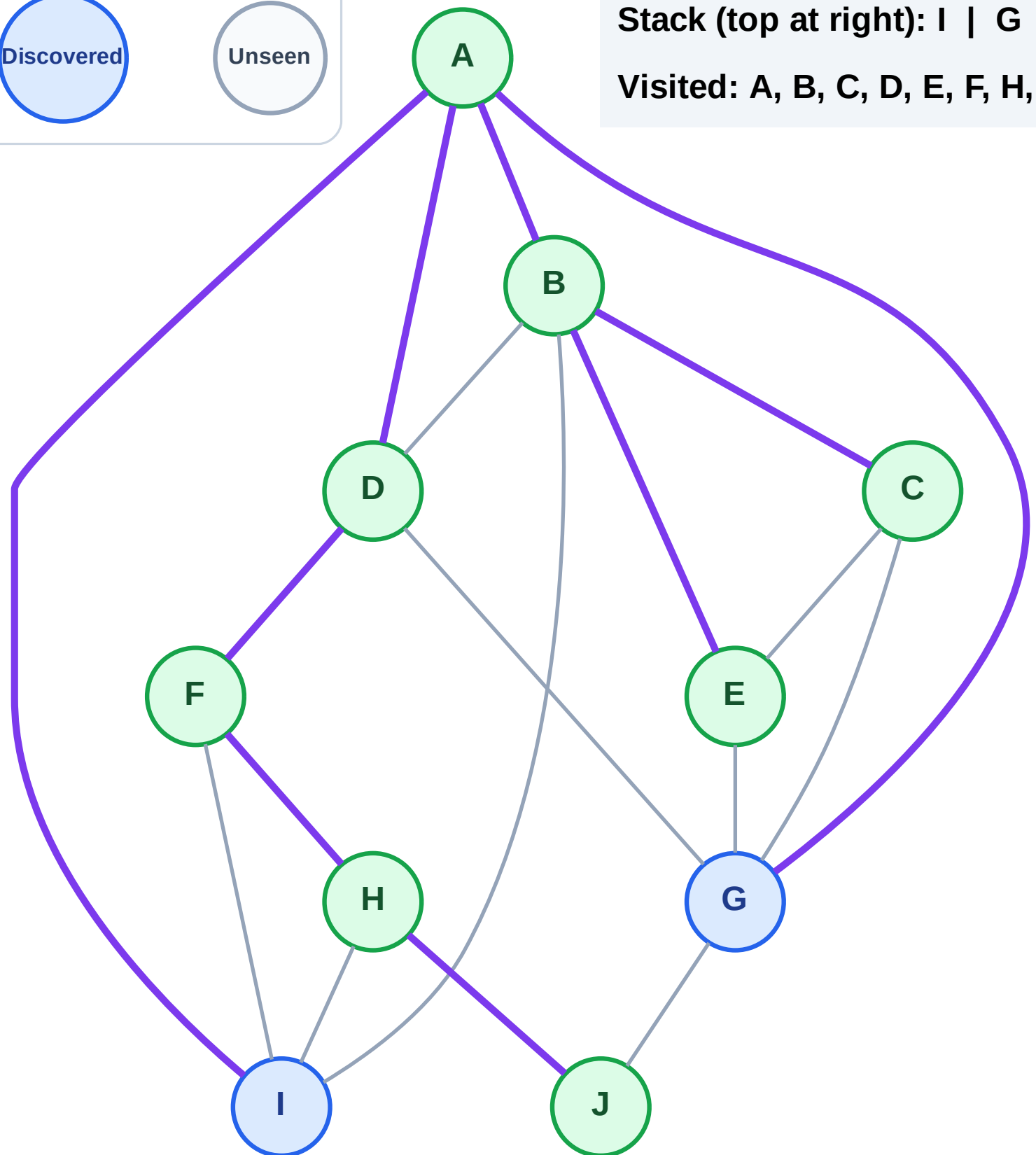
Step 17: No new neighbors from J

Legend



Stack (top at right): I | G

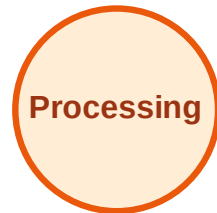
Visited: A, B, C, D, E, F, H, J



DFS Traversal

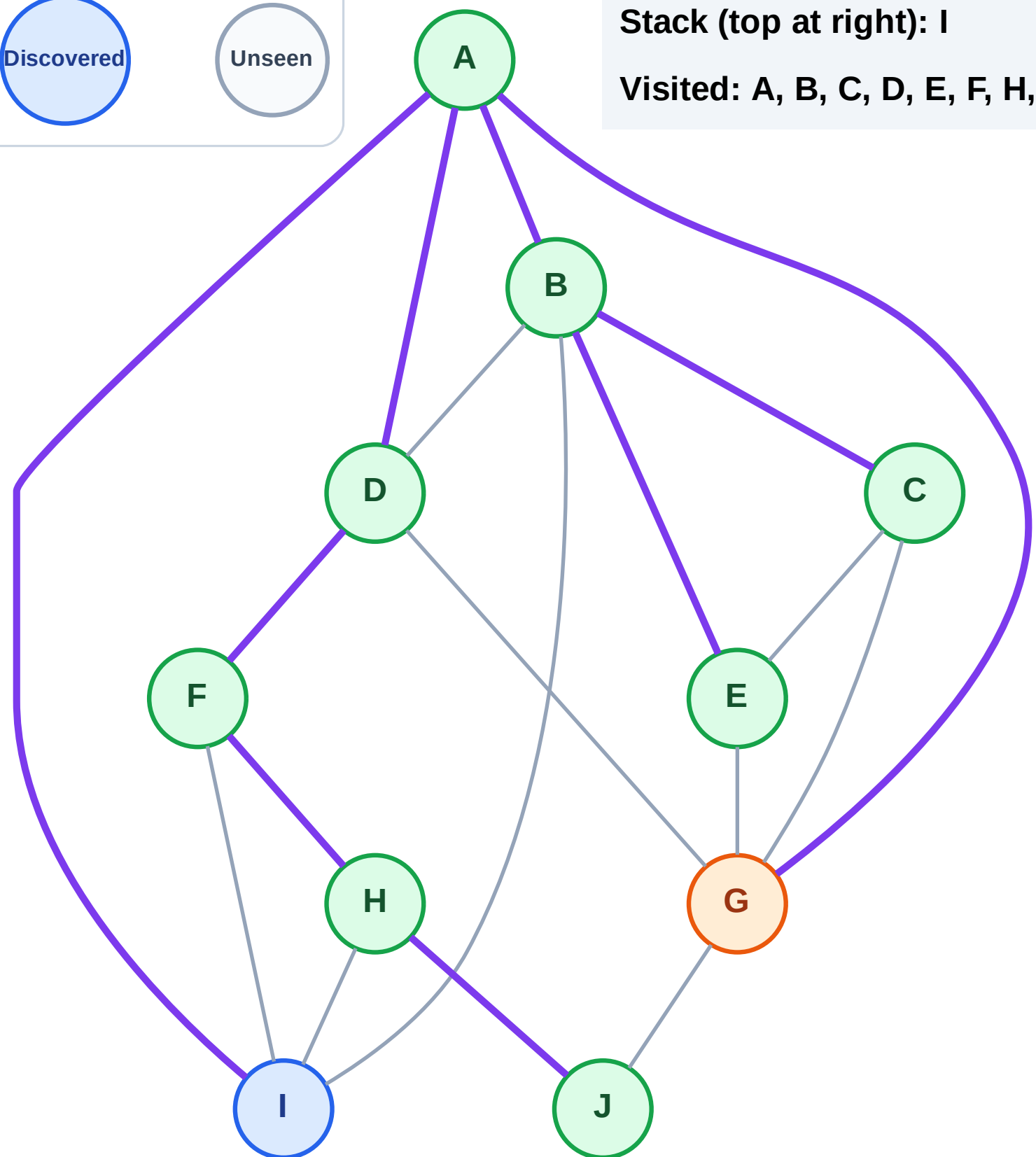
Step 18: Process node G

Legend



Stack (top at right): I

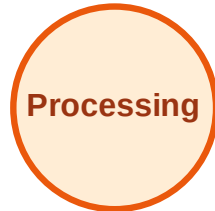
Visited: A, B, C, D, E, F, H, J



DFS Traversal

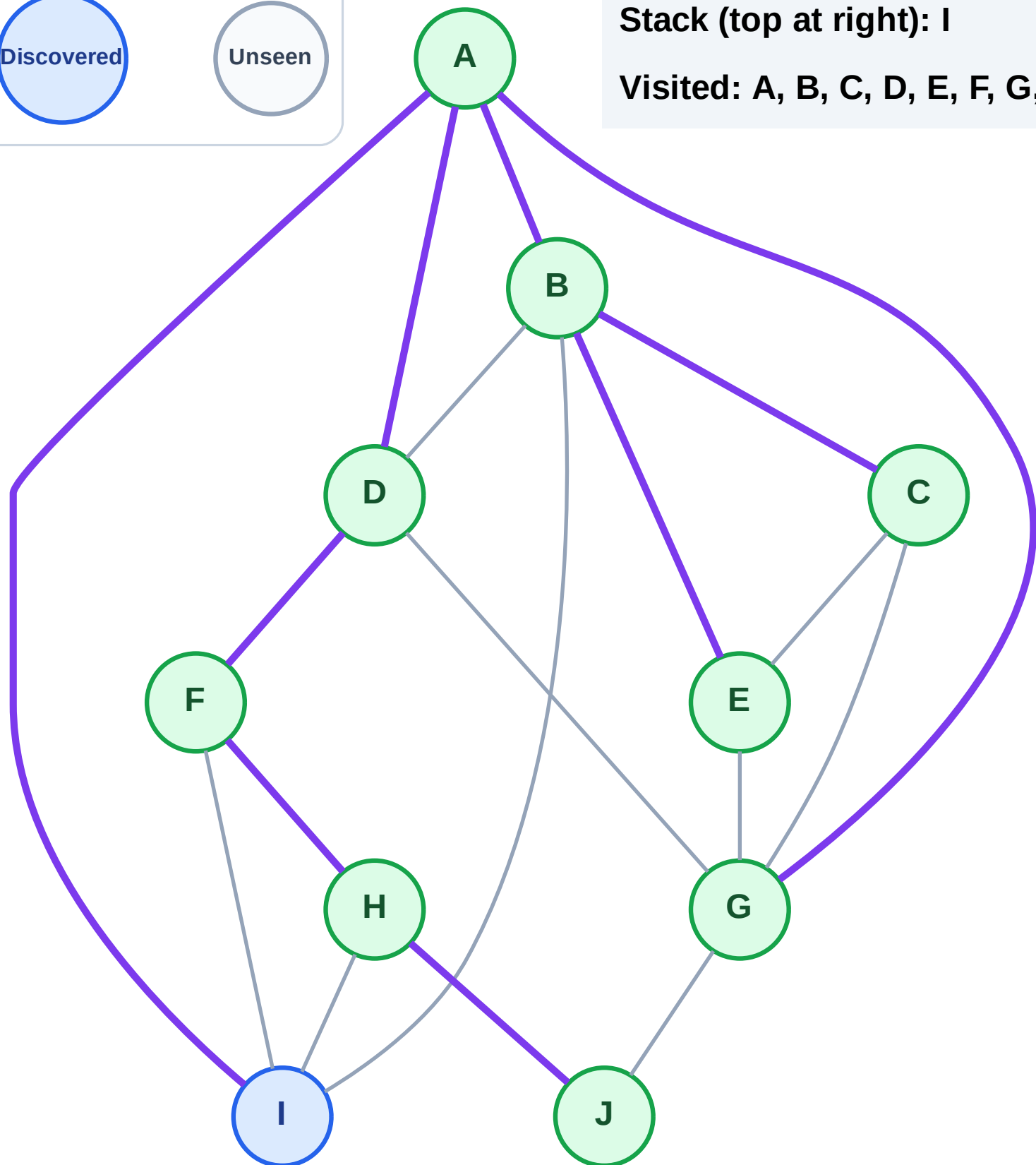
Step 19: No new neighbors from G

Legend



Stack (top at right): I

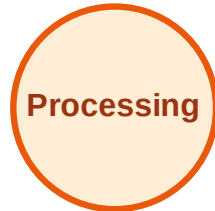
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

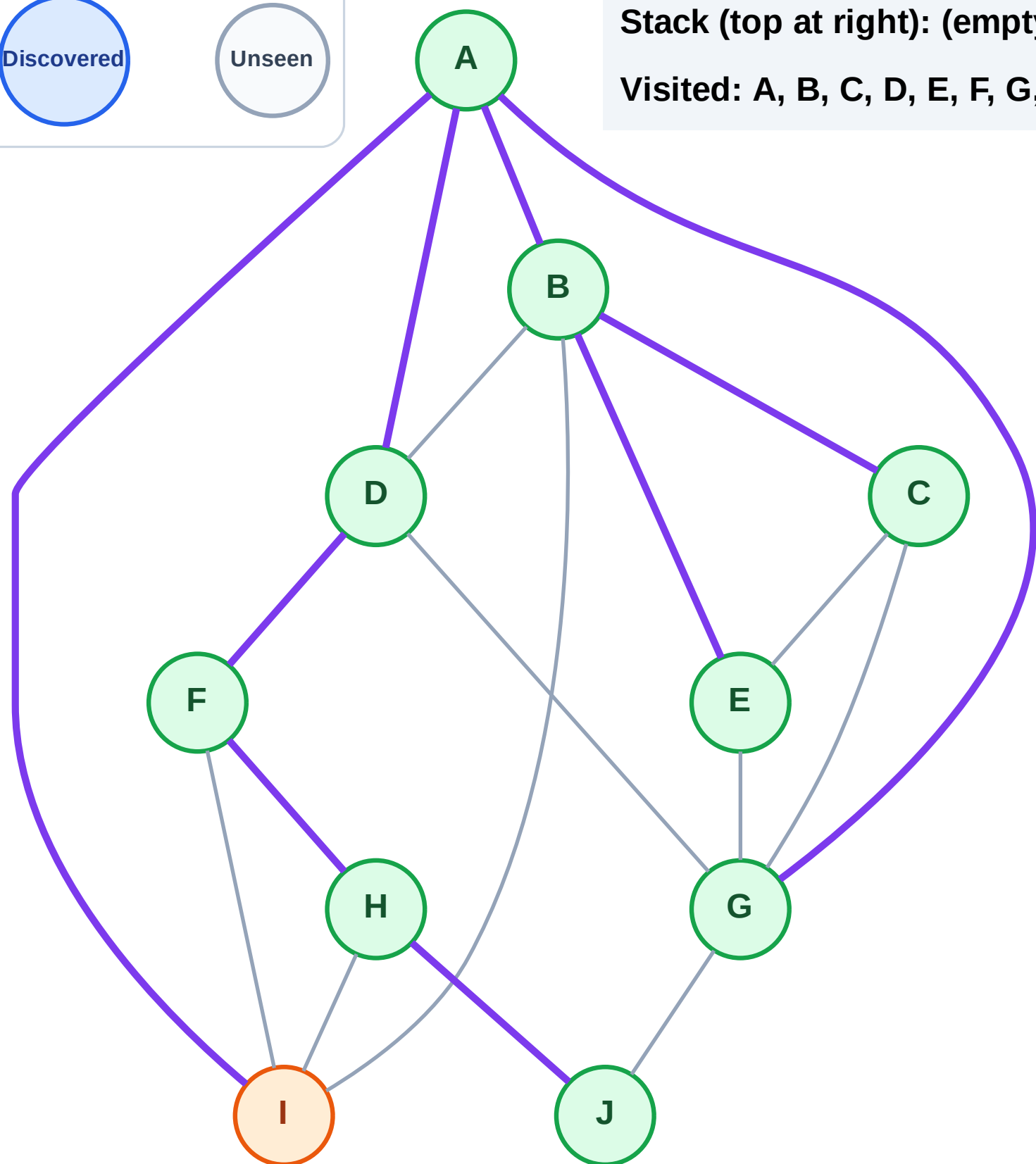
Step 20: Process node I

Legend



Stack (top at right): (empty)

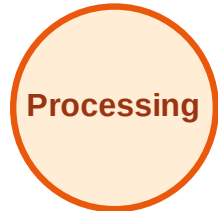
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

